AWS DataSync – Copy Data from On-Premises SMB Storage to Amazon S3

To Begin with the Lab

Summary of the Lab

In this lab, **AWS DataSync** was used to copy data from an on-premises-style SMB storage setup into **Amazon S3**. To simulate the on-premises environment, a Windows Server 2012 virtual machine was created in **VMware Workstation**, configured with a shared folder, and given test files to transfer. A DataSync agent was downloaded as an OVA, imported into VMware, assigned a static IP, and activated through the AWS console. After that, a DataSync task was created with the SMB share as the source and an S3 bucket as the destination, using an automatically generated IAM role. The task was started manually, and the execution history confirmed that multiple files were transferred successfully into the S3 bucket.

Understanding AWS DataSync

AWS DataSync is an online data transfer service that simplifies and accelerates moving data between on-premises storage systems and AWS storage services such as **Amazon S3**, **Amazon EFS**, and **Amazon FSx**. It exists so you can migrate, archive, protect, or move data without building a custom copy process. In this lab, the source was an SMB share hosted on a Windows Server virtual machine, and the destination was an S3 bucket in AWS. The benefit is that DataSync handles the transfer securely and automatically, while you focus only on the source and destination locations.

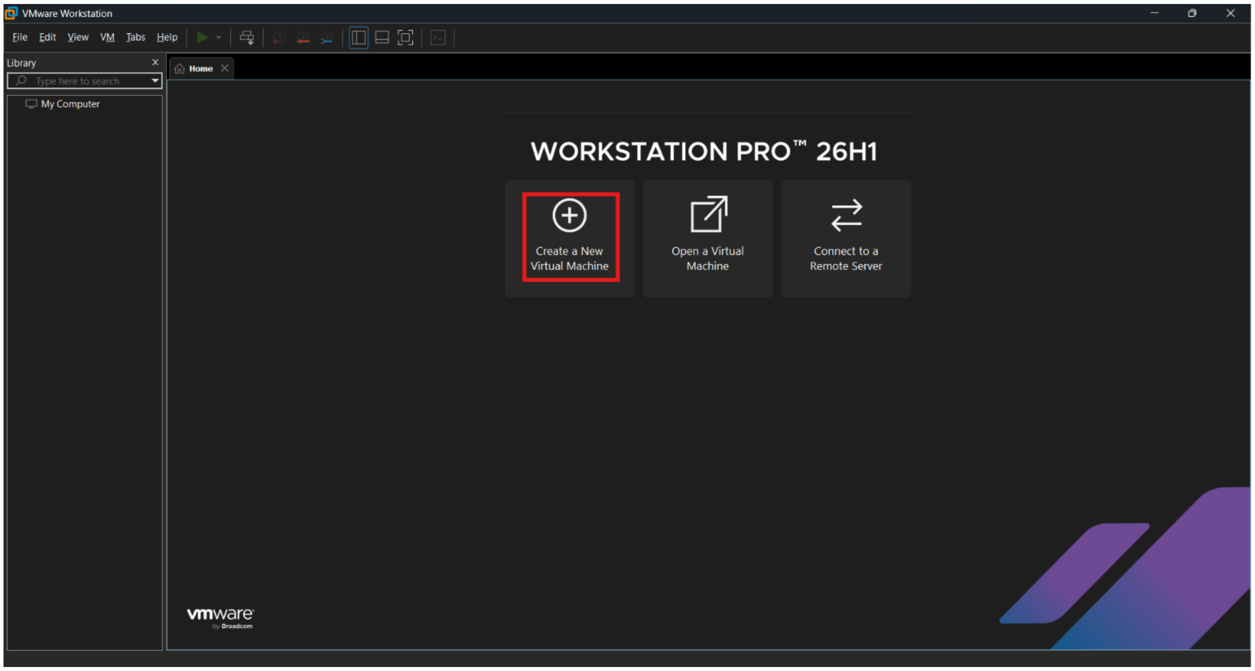
Example:

A company keeps files in a Windows SMB share named data inside a VMware-based storage server. The team wants those files to appear in an S3 bucket named on prem data for backup or processing. DataSync reads the shared folder from the source machine and copies the files into the bucket, so the same files are available in AWS without manual uploading.

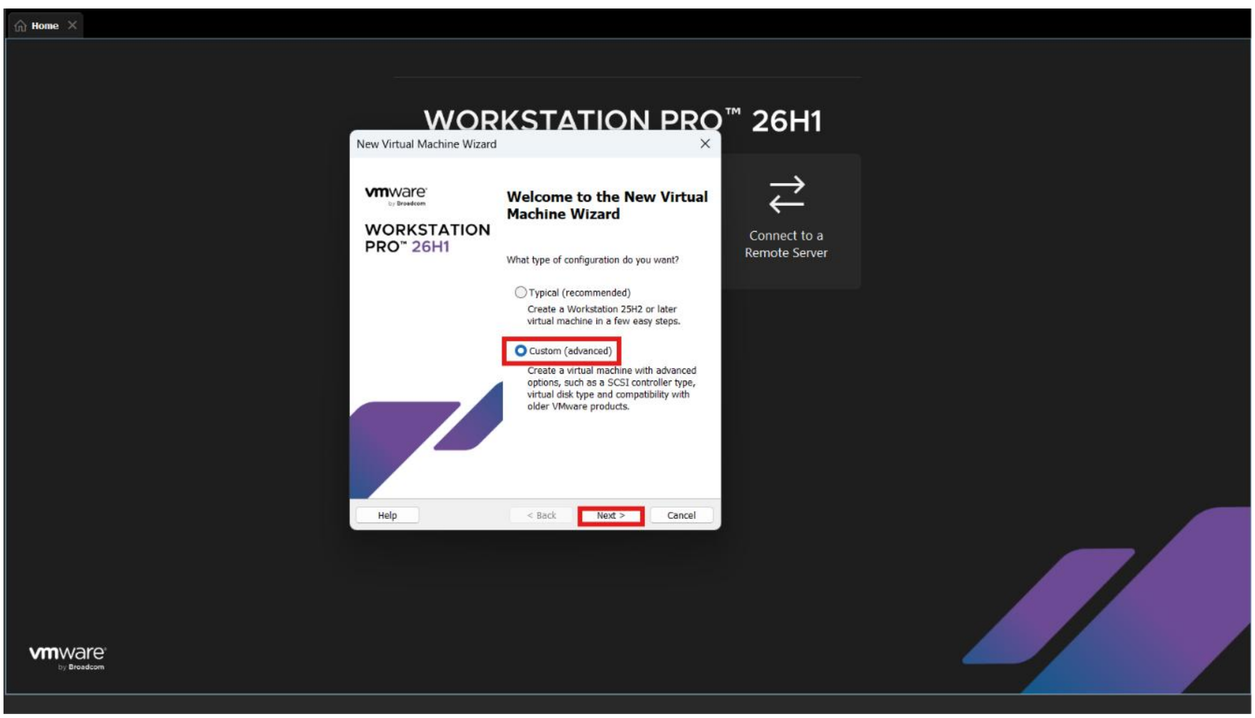
Lab Steps

Step 1 – Prepare the on-premises-style storage server

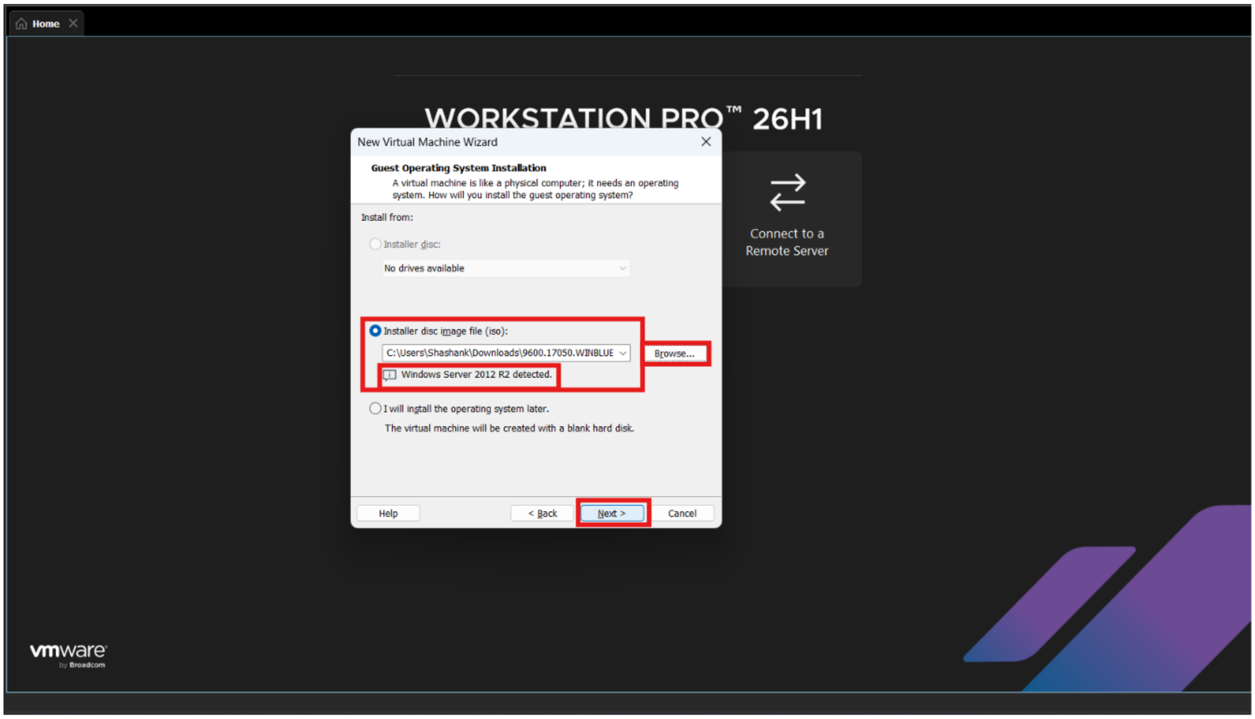
- Open **VMware Workstation**.
- Click on Create New Virtual Machine.



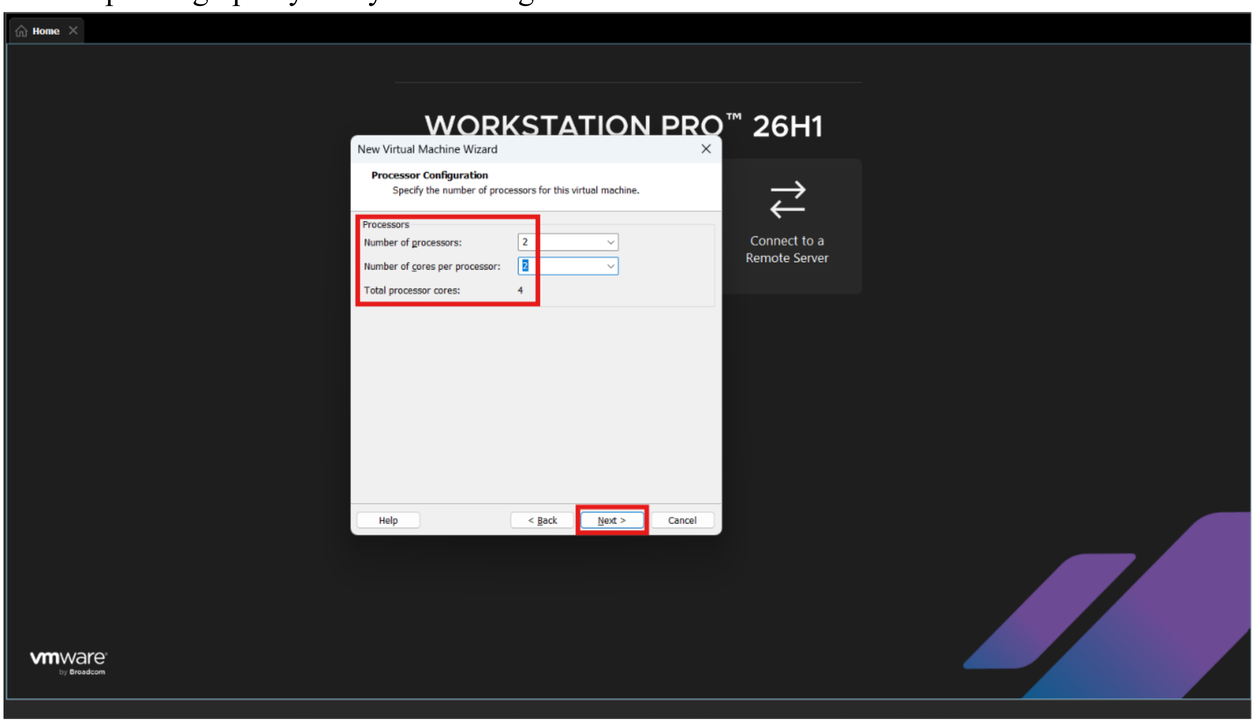
- Create a new virtual machine using **Windows Server 2012** ISO.
- If you don't have the Windows ISO you can download it through its official site.
- For VMWare you have to make an account on their official site to download VMWare Workstation Pro.
- Now choose **Custom** and Click **Next**.



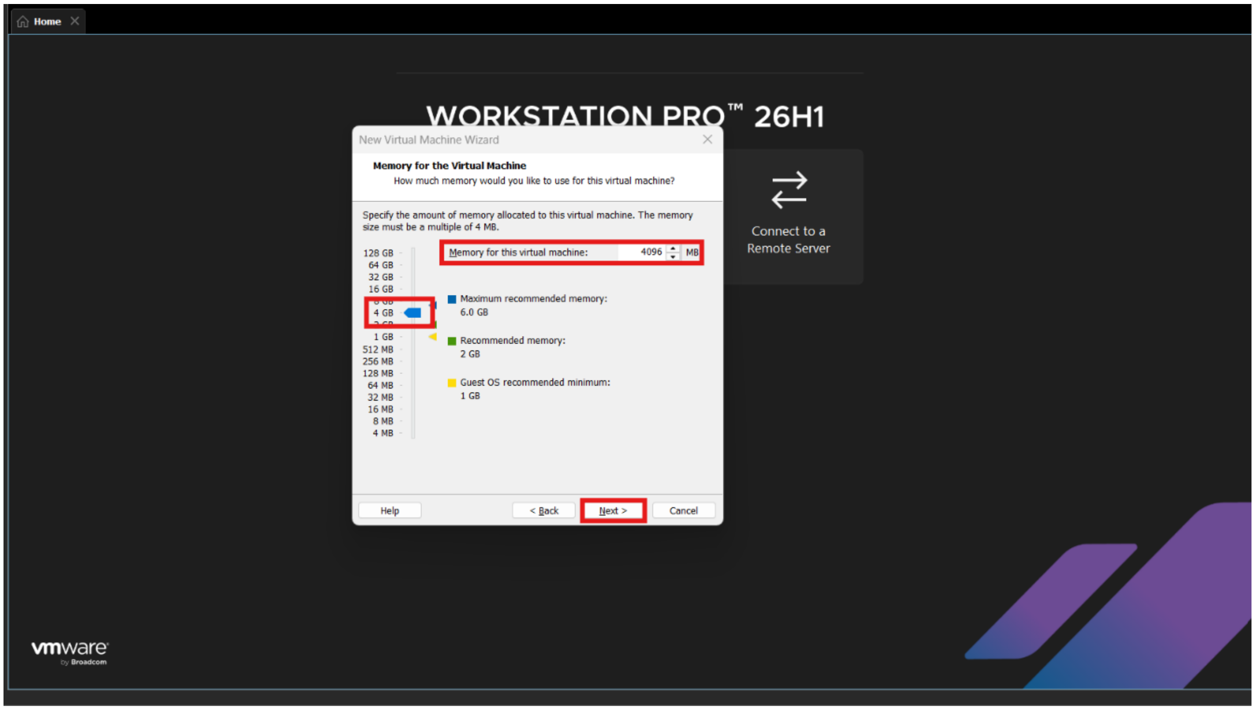
- Choose **Windows 2012 ISO** we have downloaded earlier by clicking on **Browse** and click on **Next**.



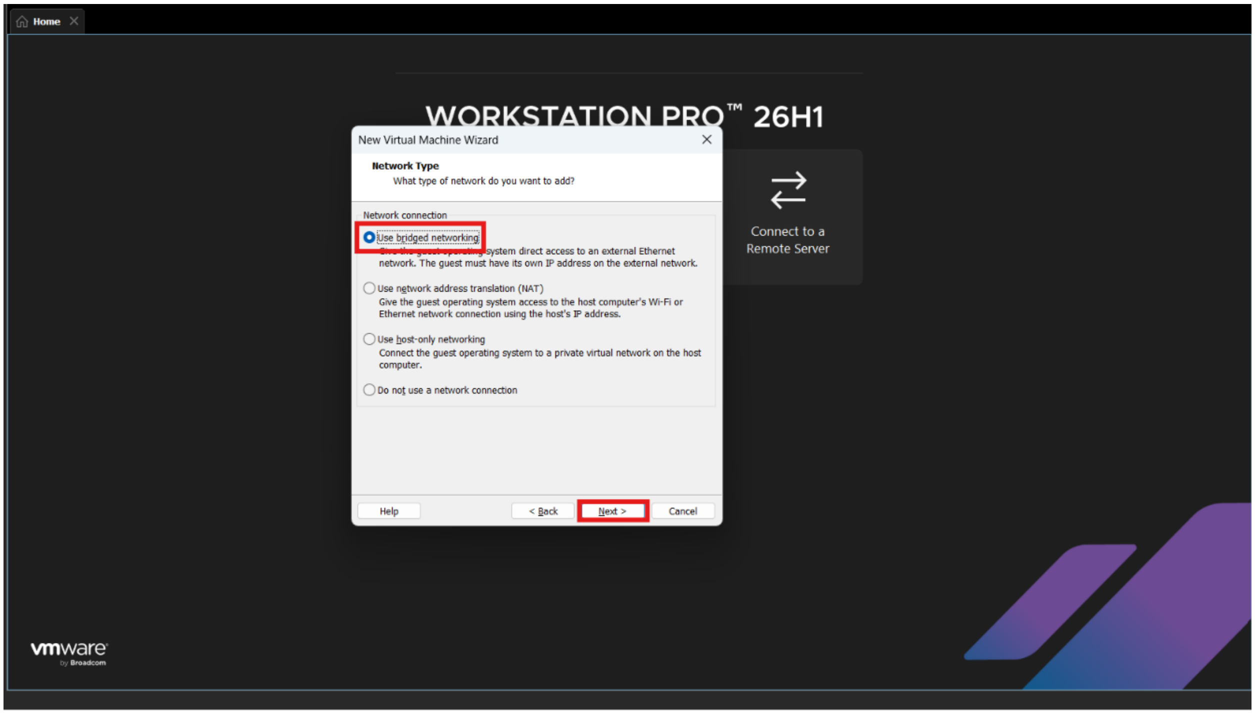
- Now I will assign **4 CPU** based on my CPU Configuration, but you can choose minimum of 2 depending upon your system configurations.



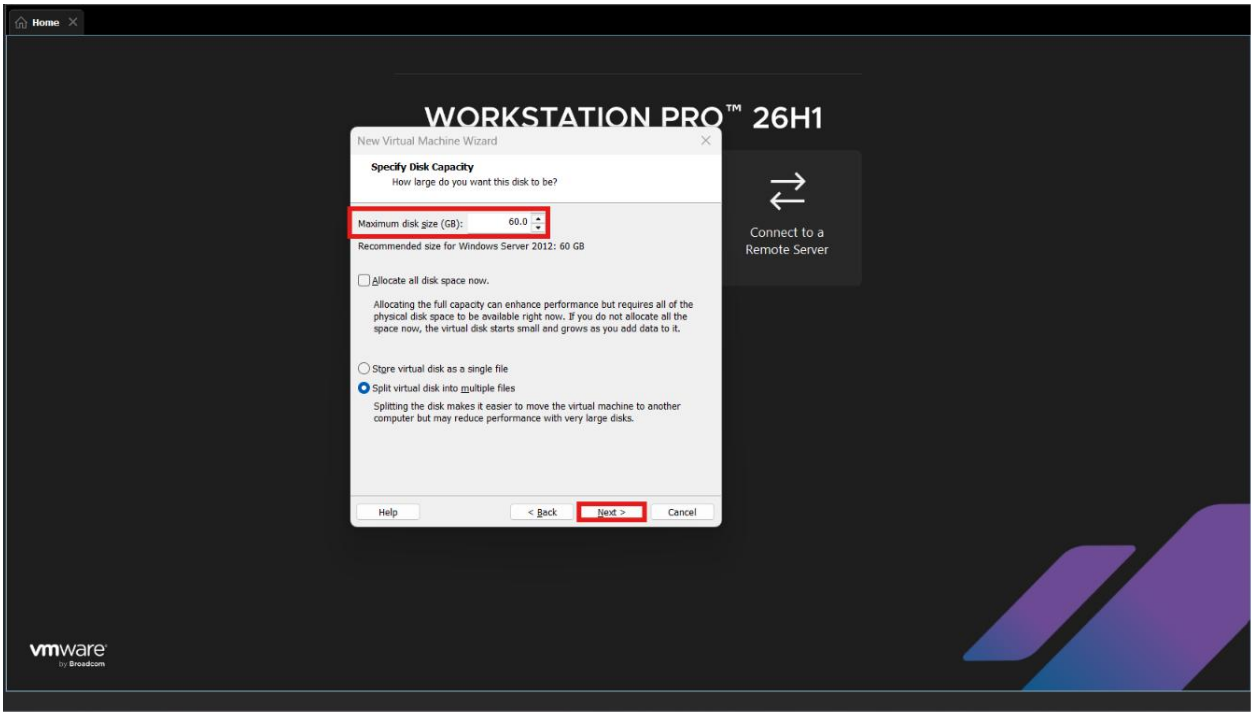
- Now provide at least **4 GB** of RAM for the machine.



- Choose **Bridged networking** and click **Next**.



- Leave rest options as default and keep clicking on **Next.**
- Choose **Maximum disk size** as **60GB** depending on your pc you can configure more as well.

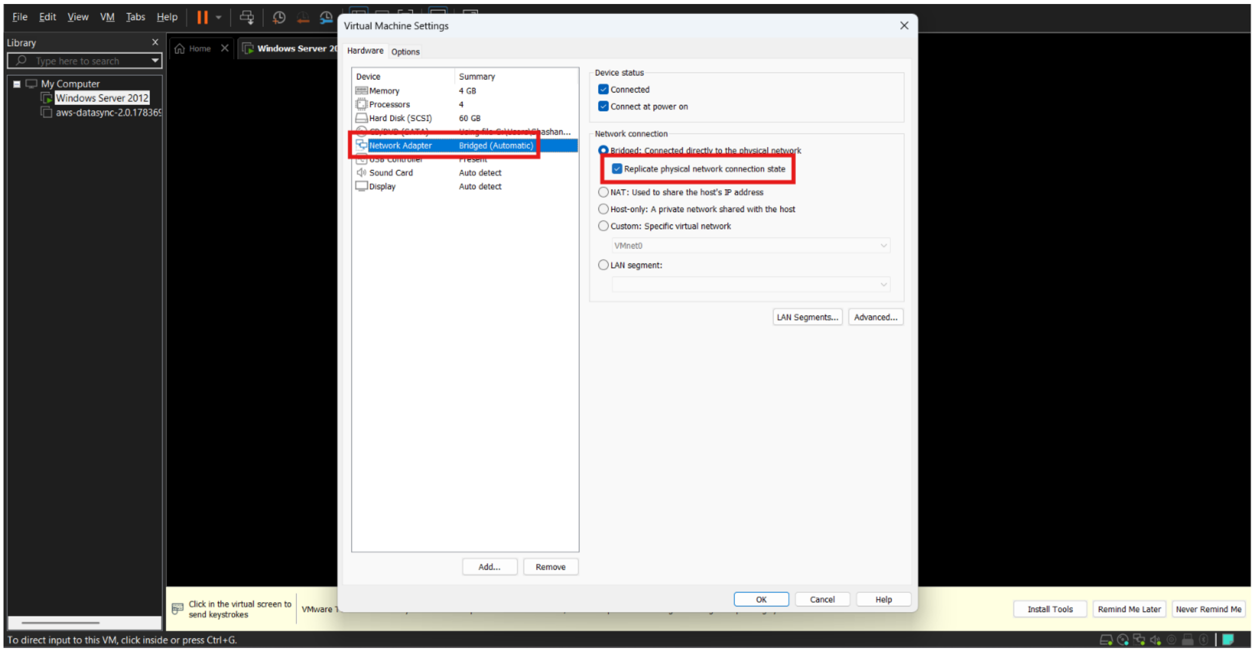


- Finish the VM creation.

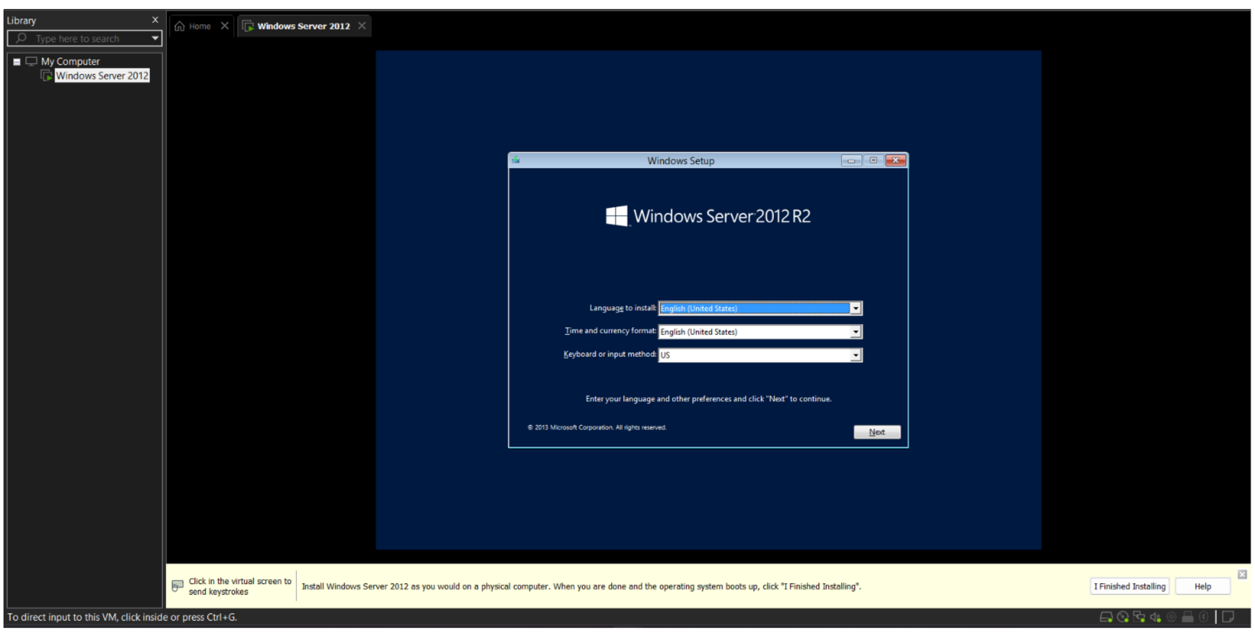
Step 2 – Install and configure Windows Server

- Right-click on **Virtual Machine** and click on **Settings**.

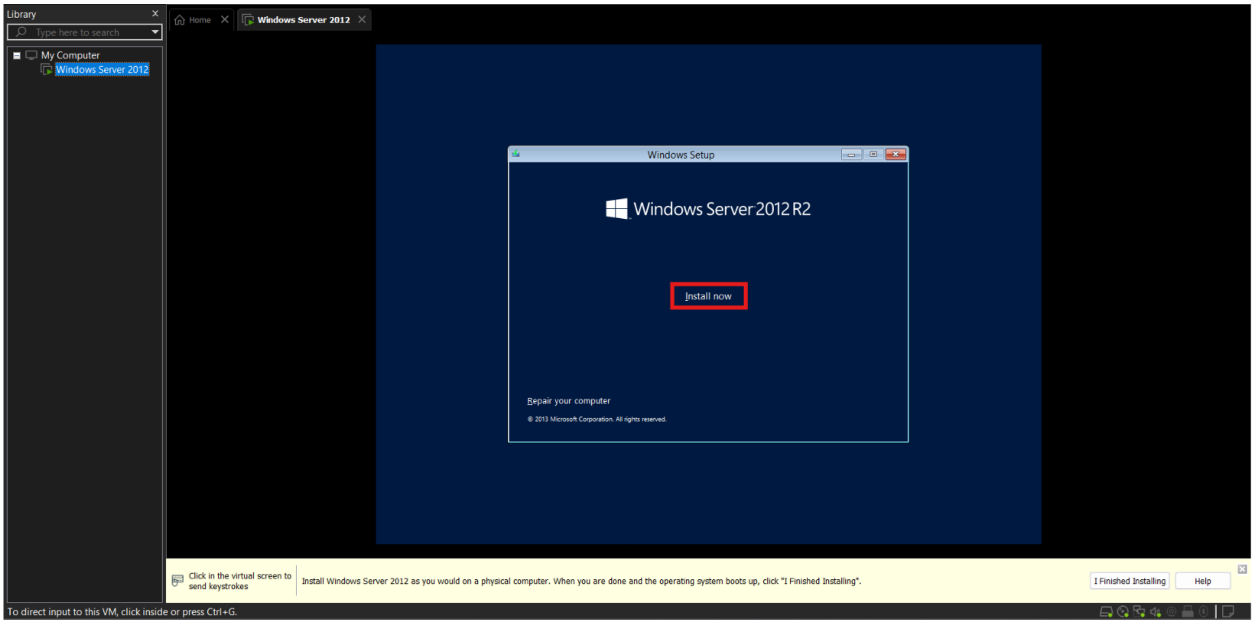
- Now Click on **Network Adapter** and Enable **Replicate physical network state connection**.



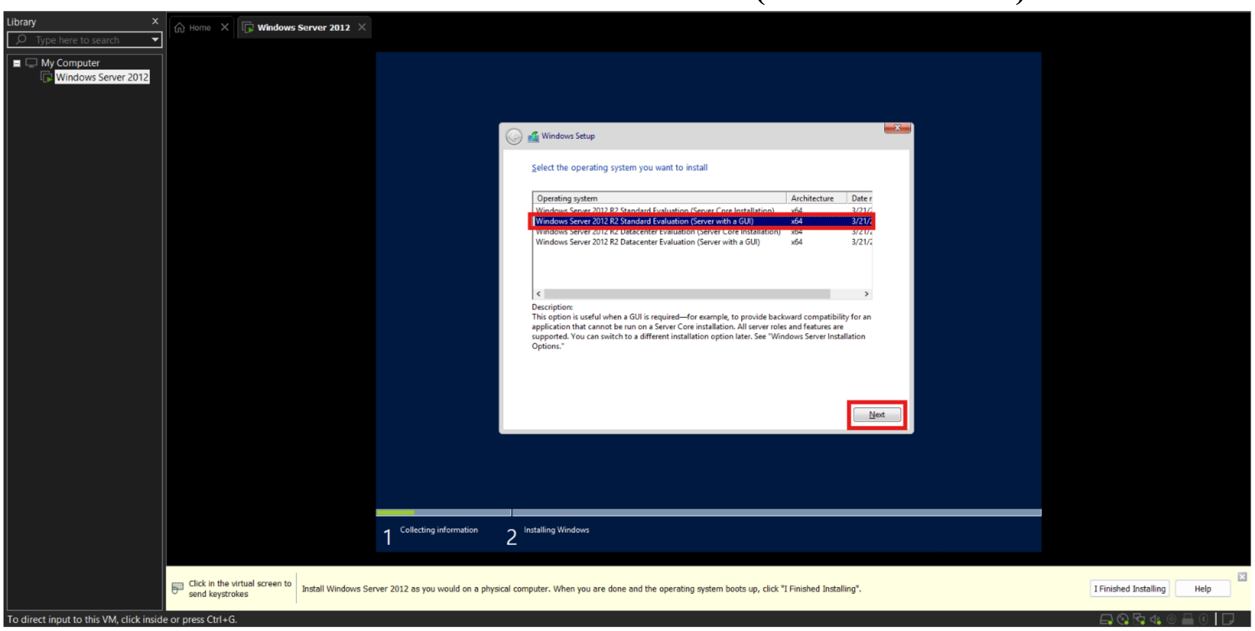
- Power on the virtual machine.
- Install **Windows Server 2012 Standard with GUI**.



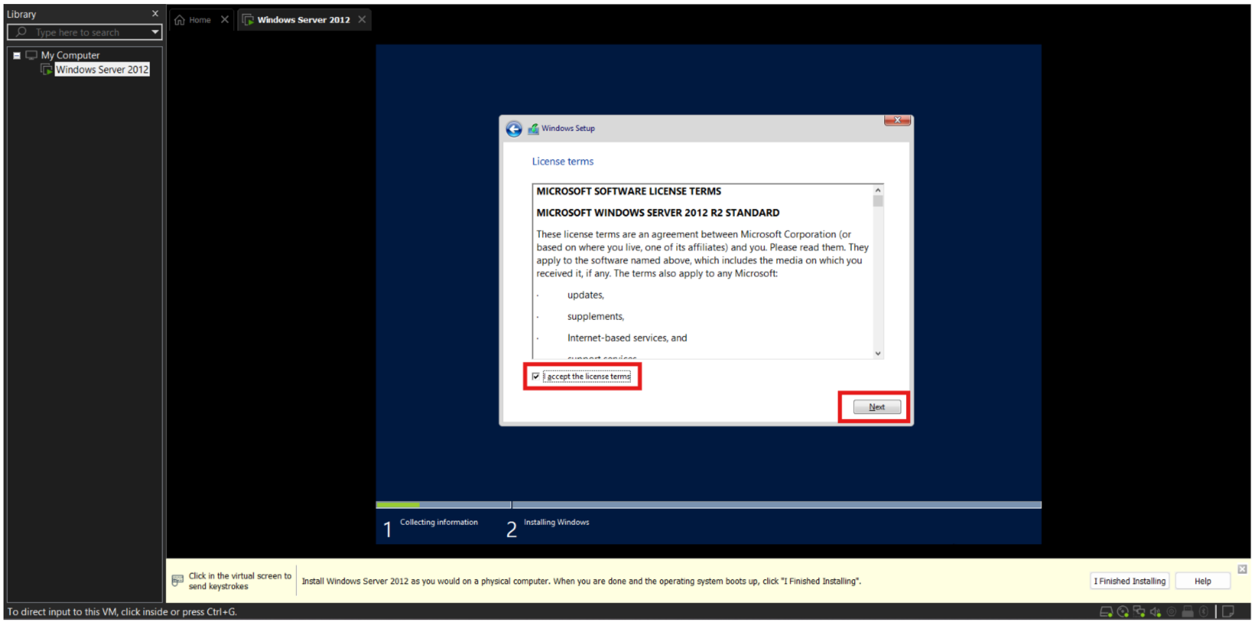
- Click **Next** and choose the **Install Now**.



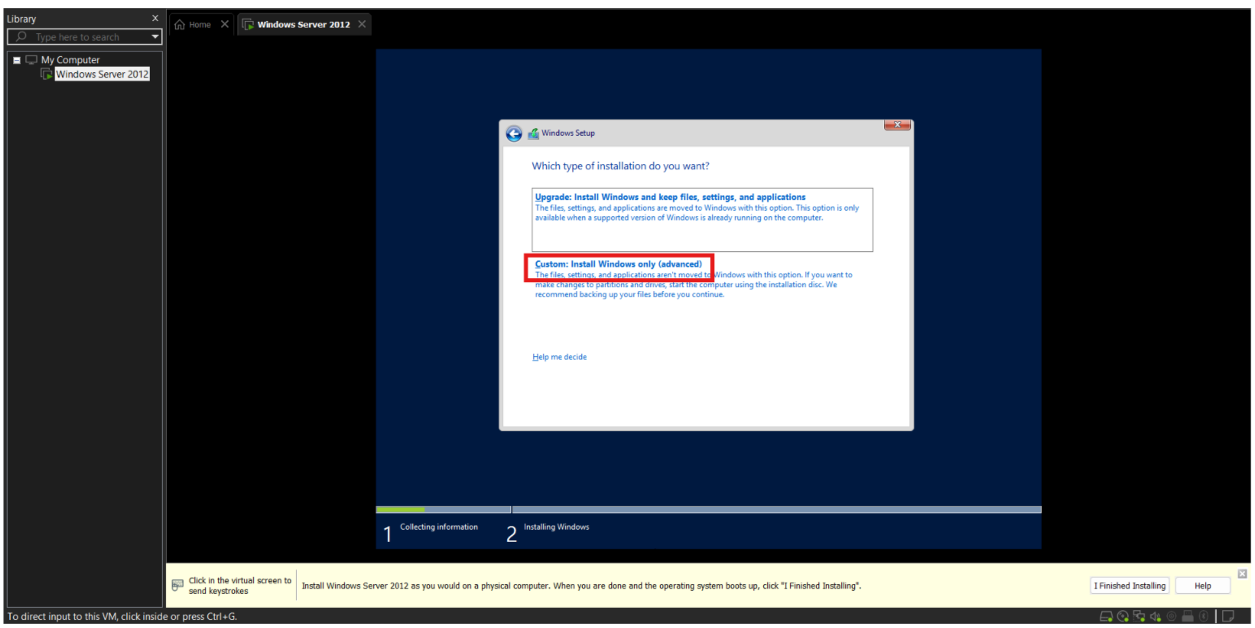
- Choose **Windows Server 2012 Standard Evaluation (Server with a GUI)** and click **Next**.



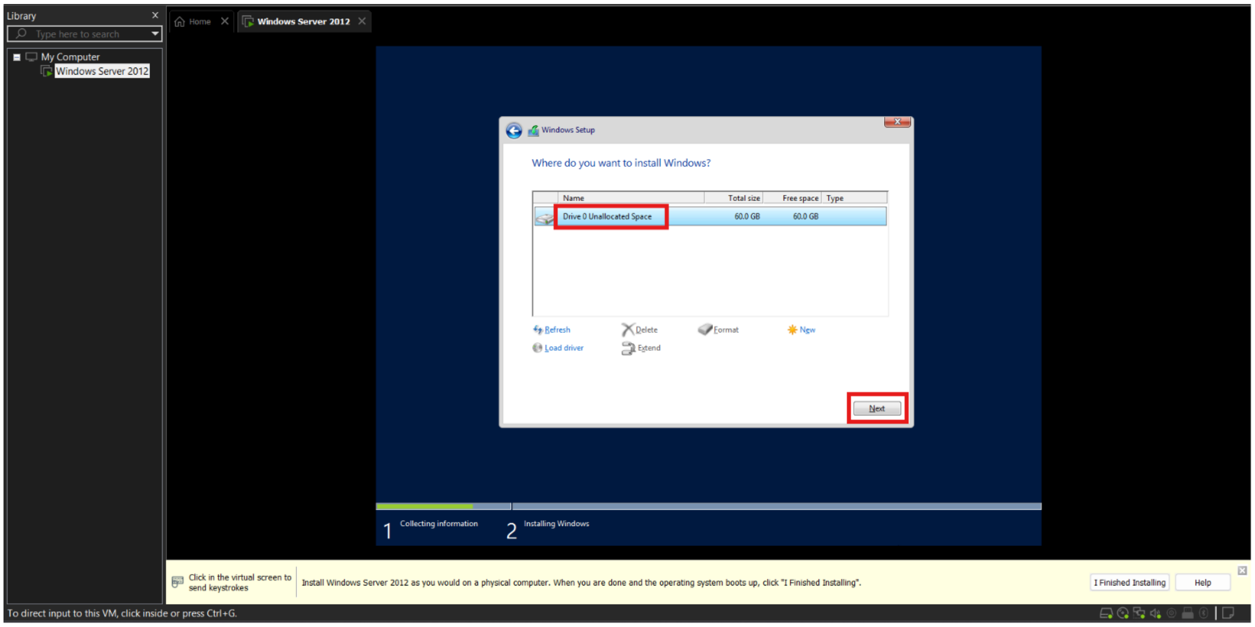
- **Accept License** and click **Next**.



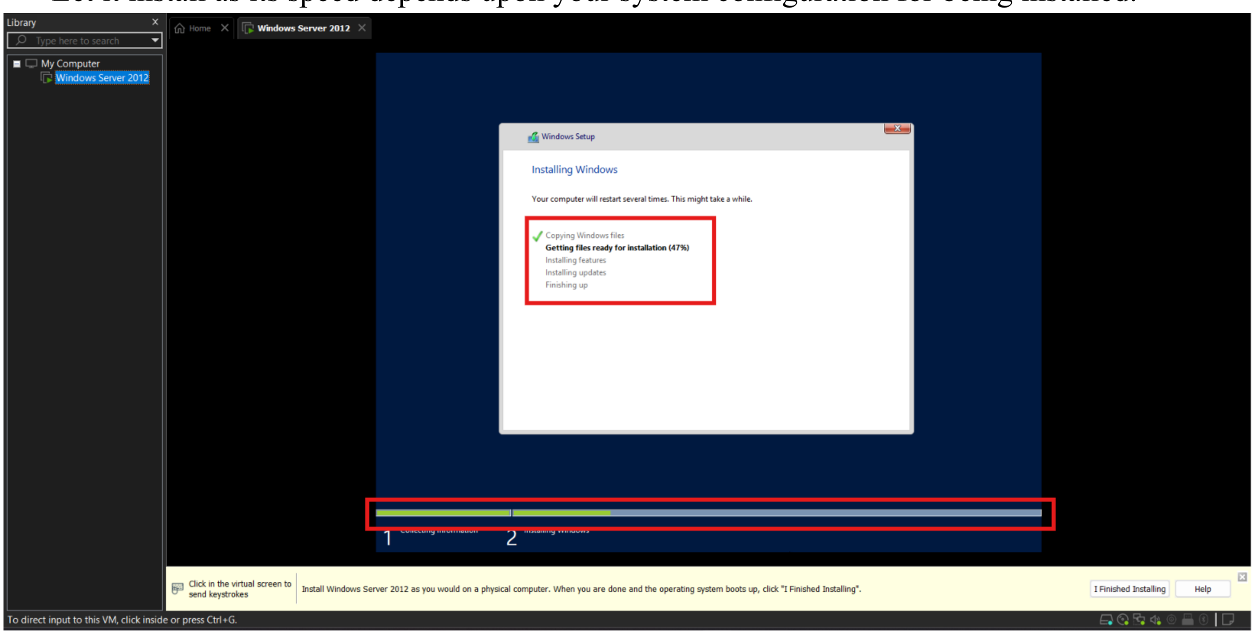
- Click on **Custom**.



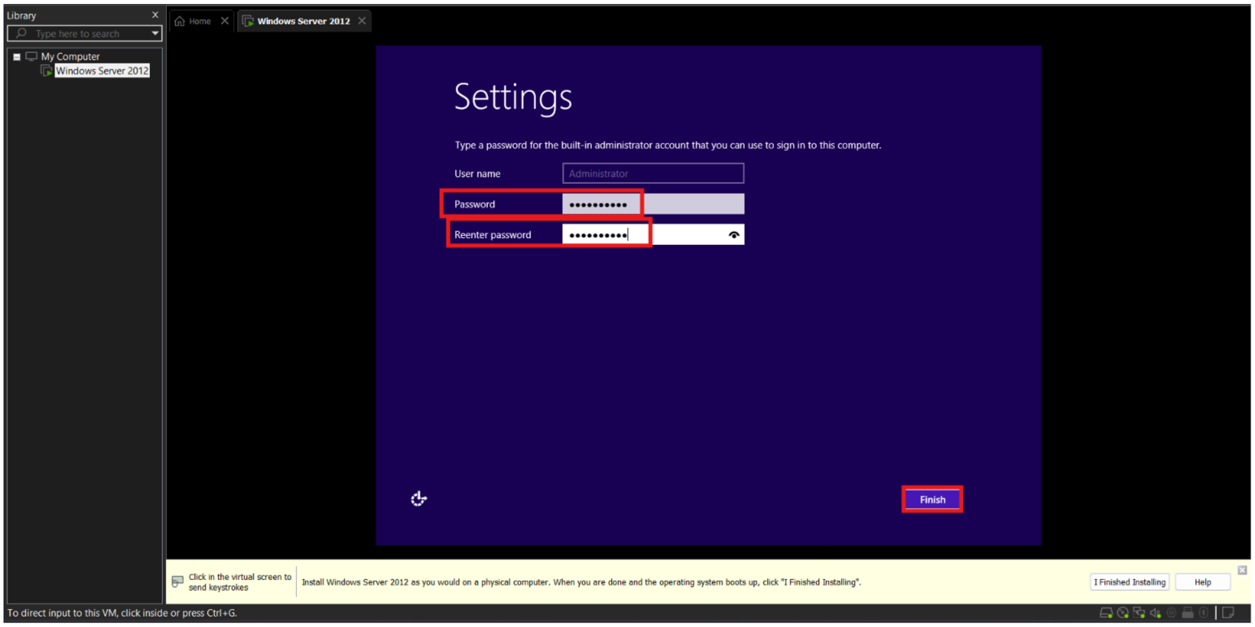
- Now click on the **Drive** and then proceed with installation by clicking on **Next**.



- Let it install as its speed depends upon your system configuration for being installed.

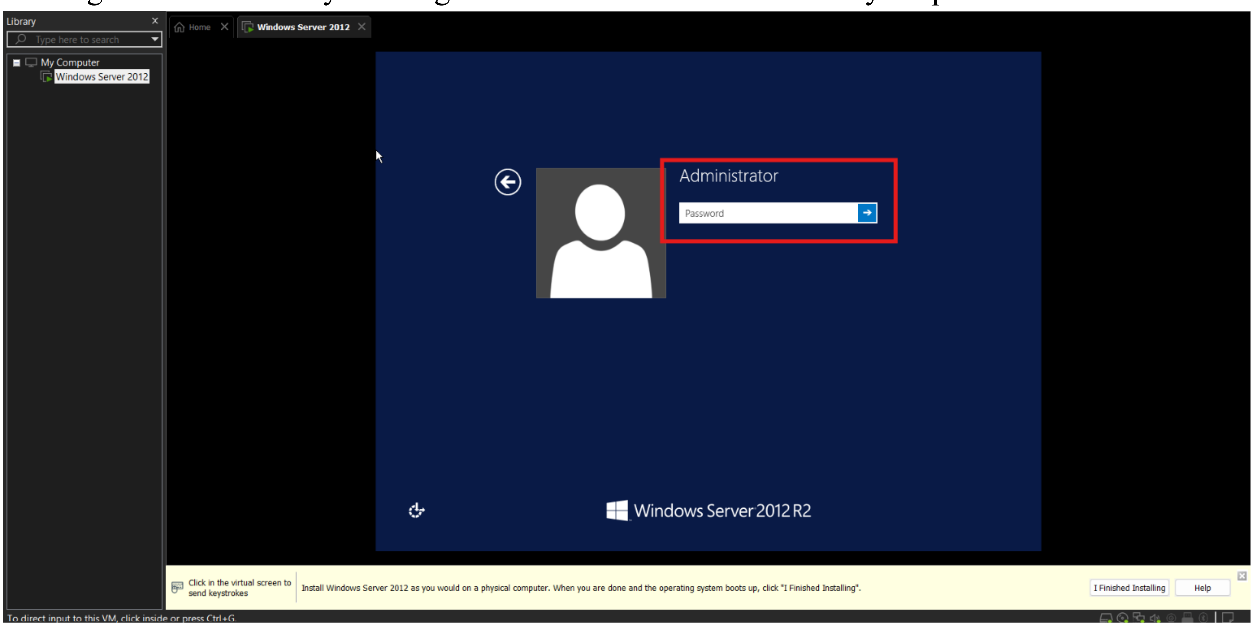


- Set the administrator password and click **Finish**.

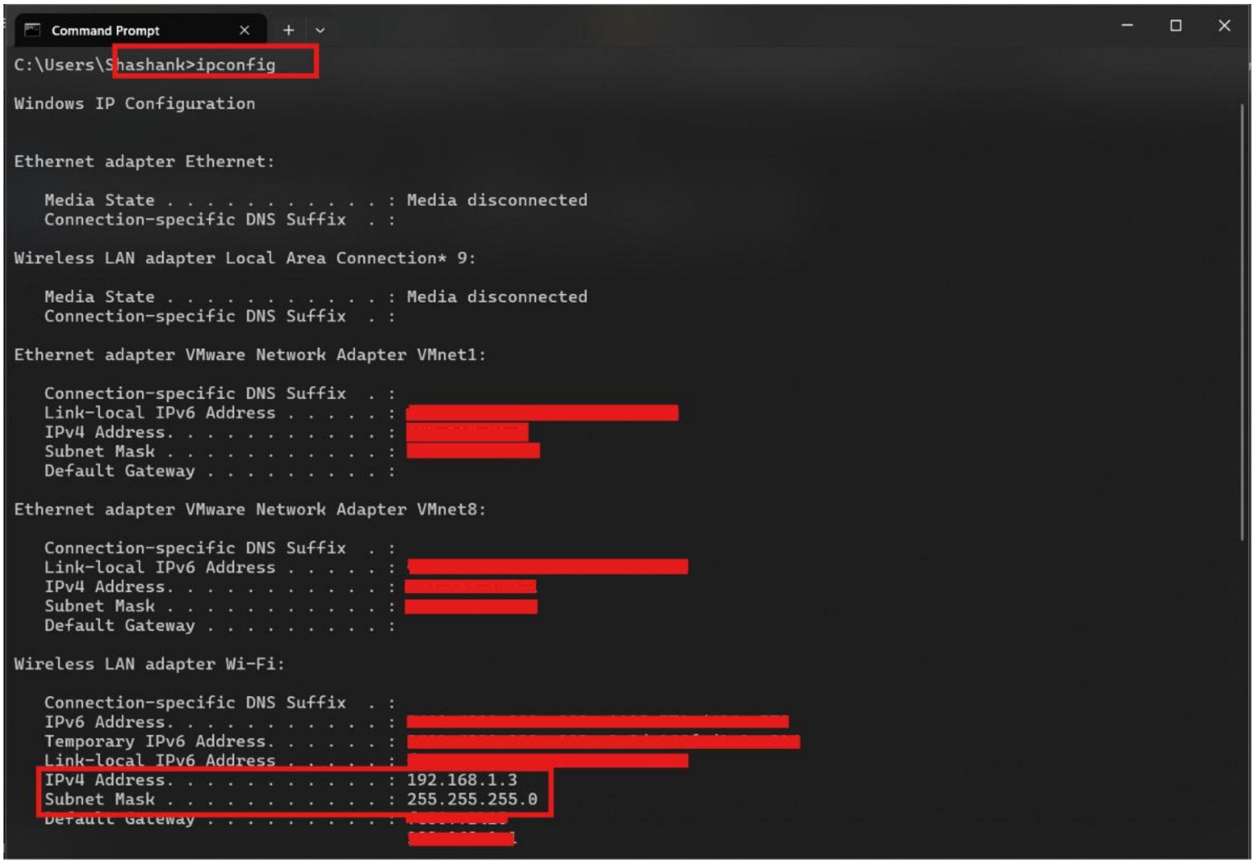


Step 3 – Configure IP Settings

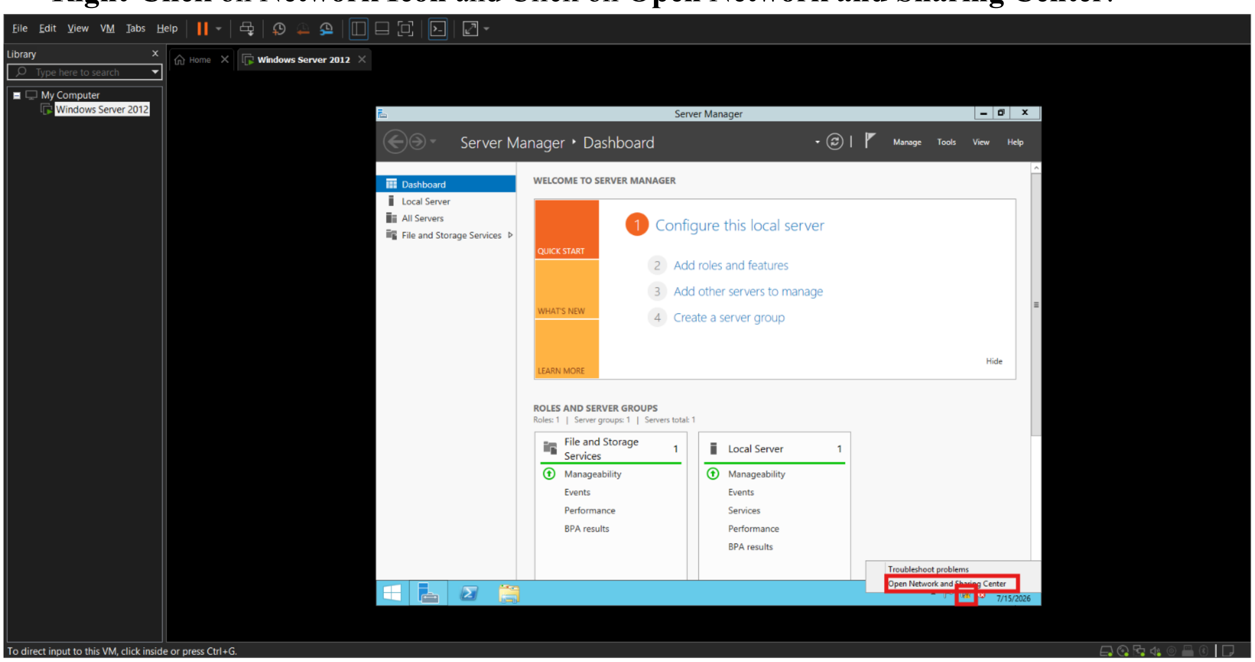
- Login to the Server by clicking Ctrl+Alt+Delete and then enter your password.



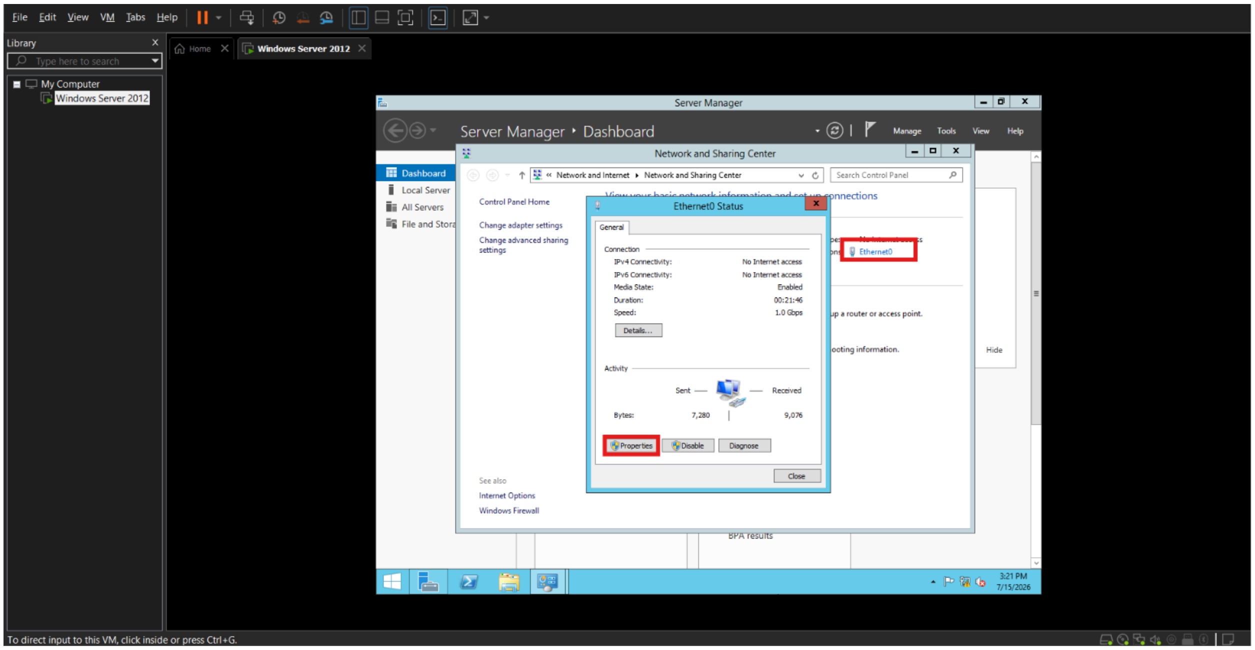
- Now open **Command Prompt** on your main computer and type **ipconfig** to get your IP Address.



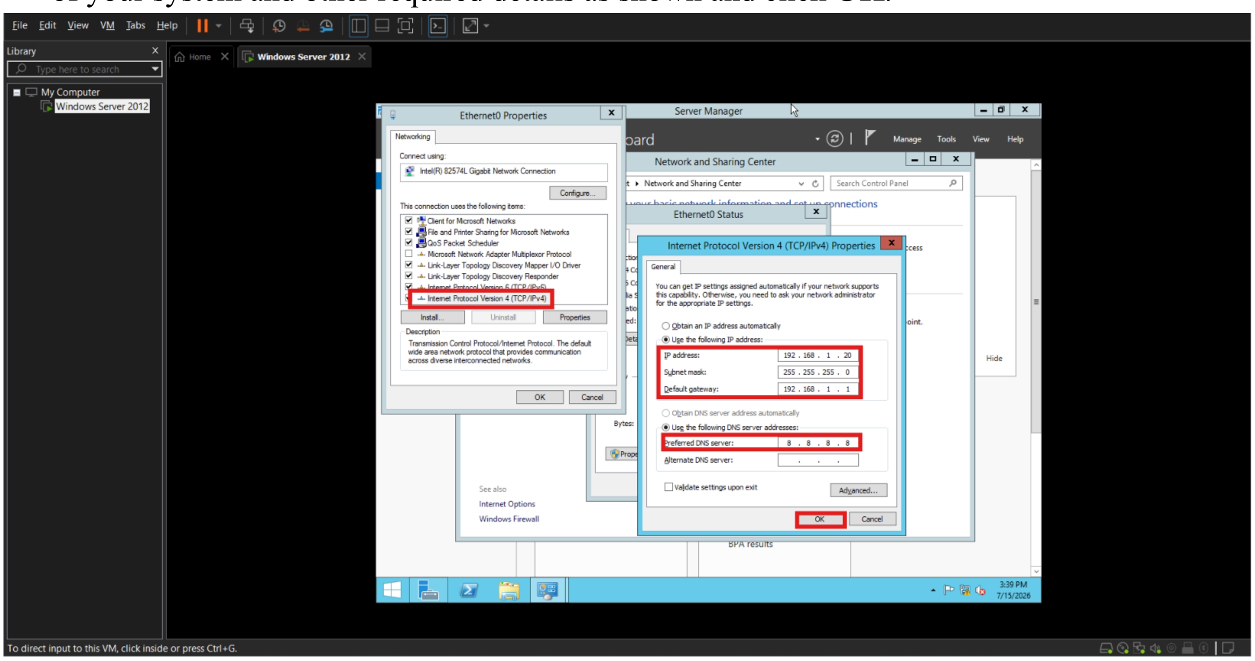
- Now we will configure the same series IP Address on the Virtual Machine so for that we will **Right-Click** on **Network Icon** and Click on **Open Network and Sharing Center**.



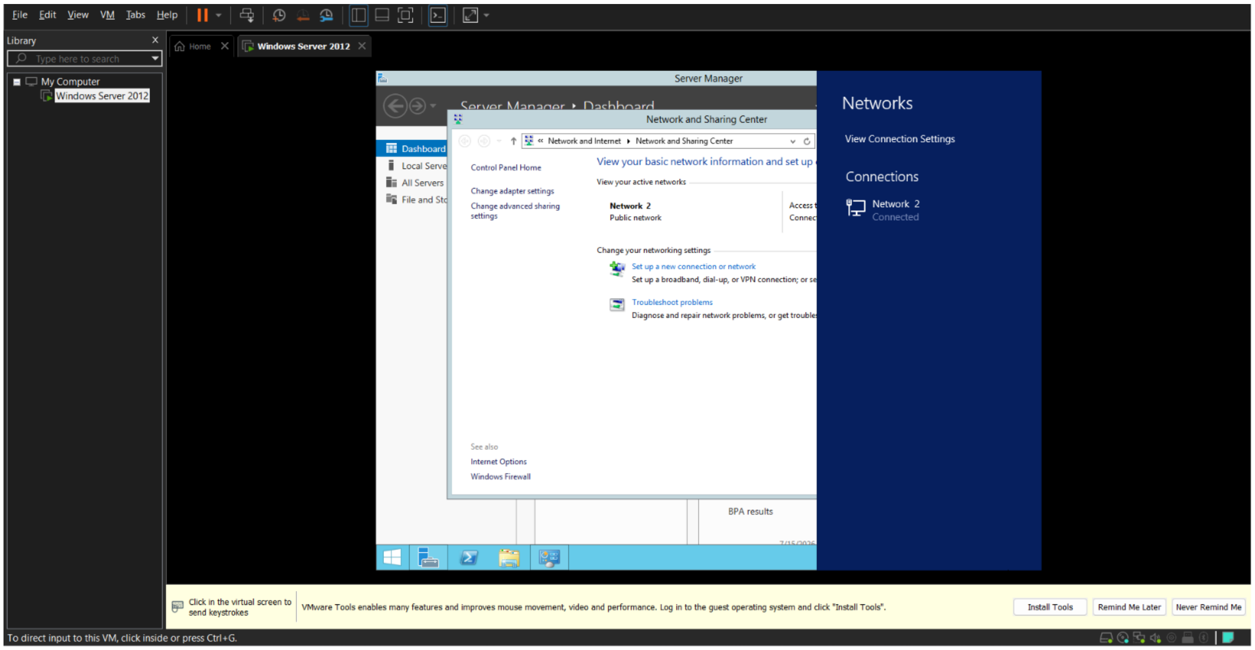
- Now click on **Ethernet** and then click on **Properties**.



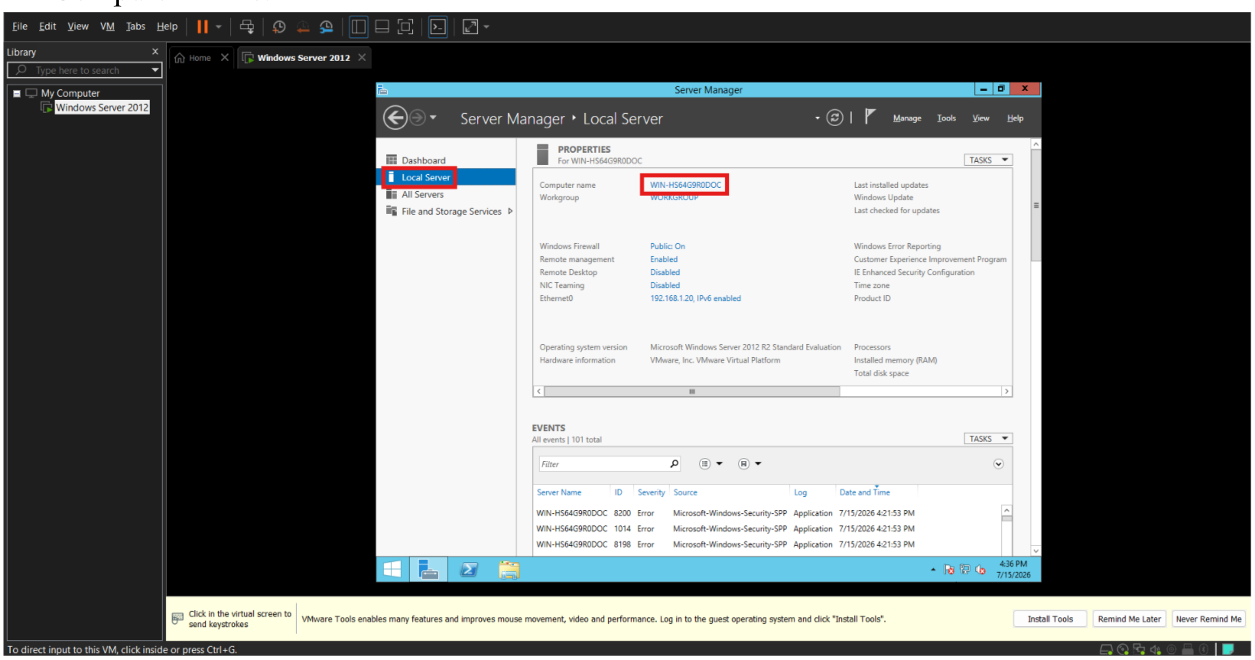
- Now Open **properties** of **Internet Protocol Version 4 (TCP/IPv4)** and enter the IP Address of your system and other required details as shown and click **OK**.



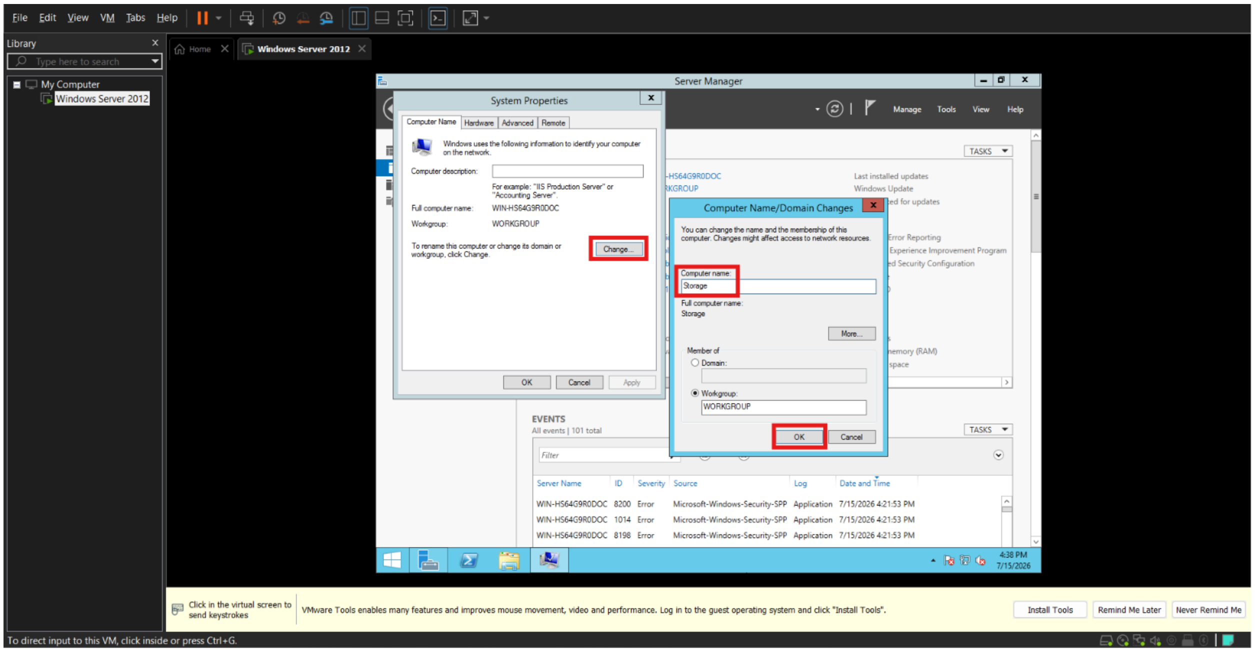
- Click **Yes** for this network prompt and you will be connected to **Internet**.



- Change the name of the computer by opening the Server Manager → Local Server → Computer Name.



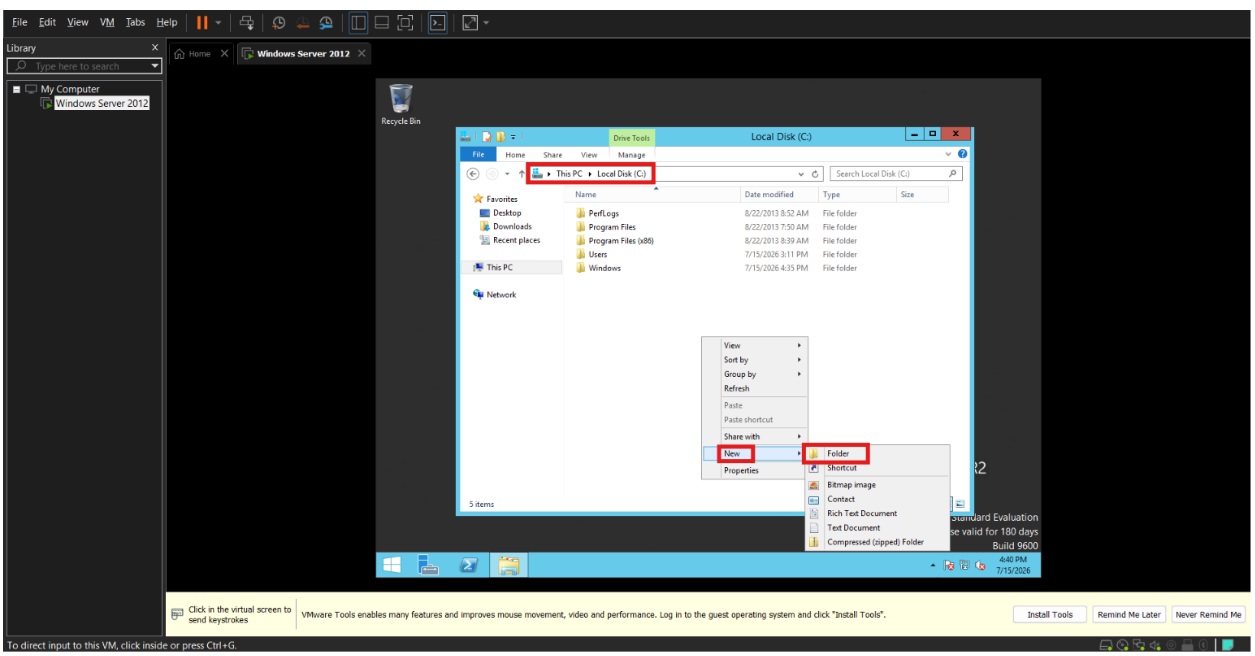
- I will rename it as **Storage** and then **Restart Later**.



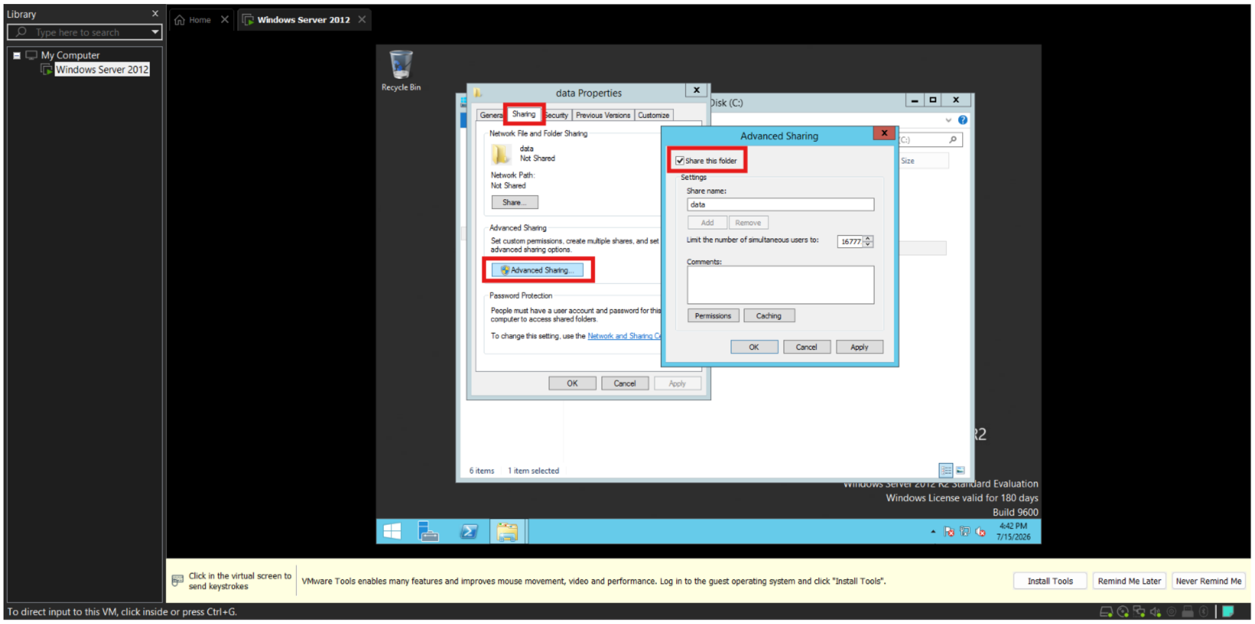
- Create a folder named data.

Step 3 – Create the shared folder

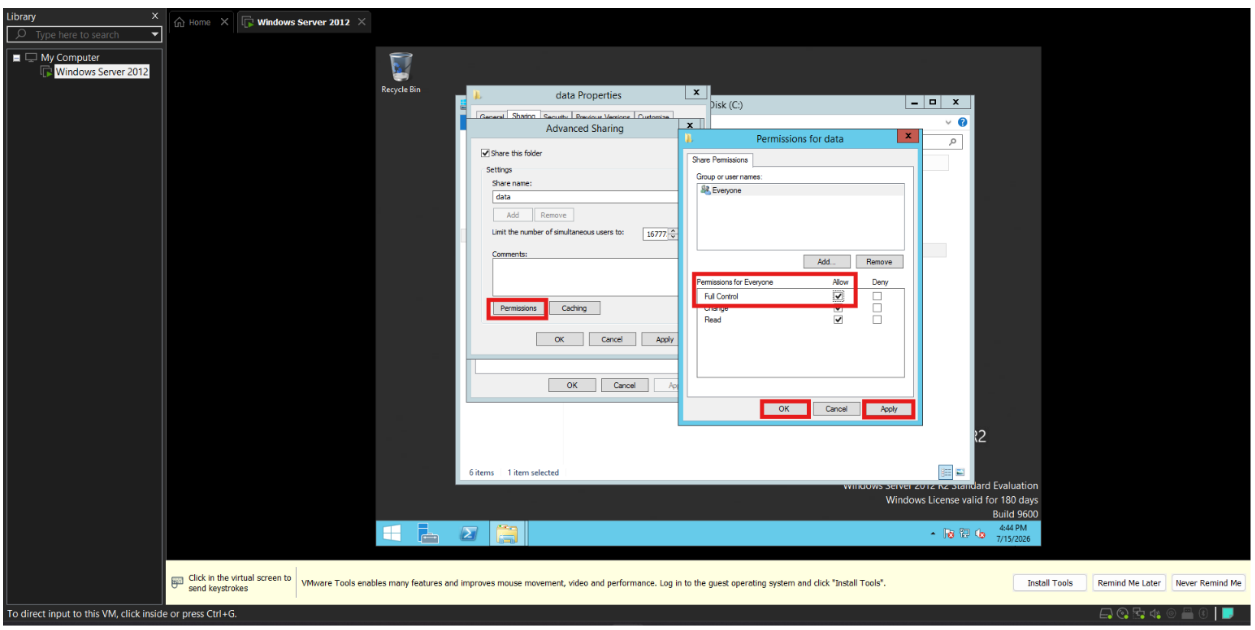
- Open the C: drive.
- Create a folder named **data**.



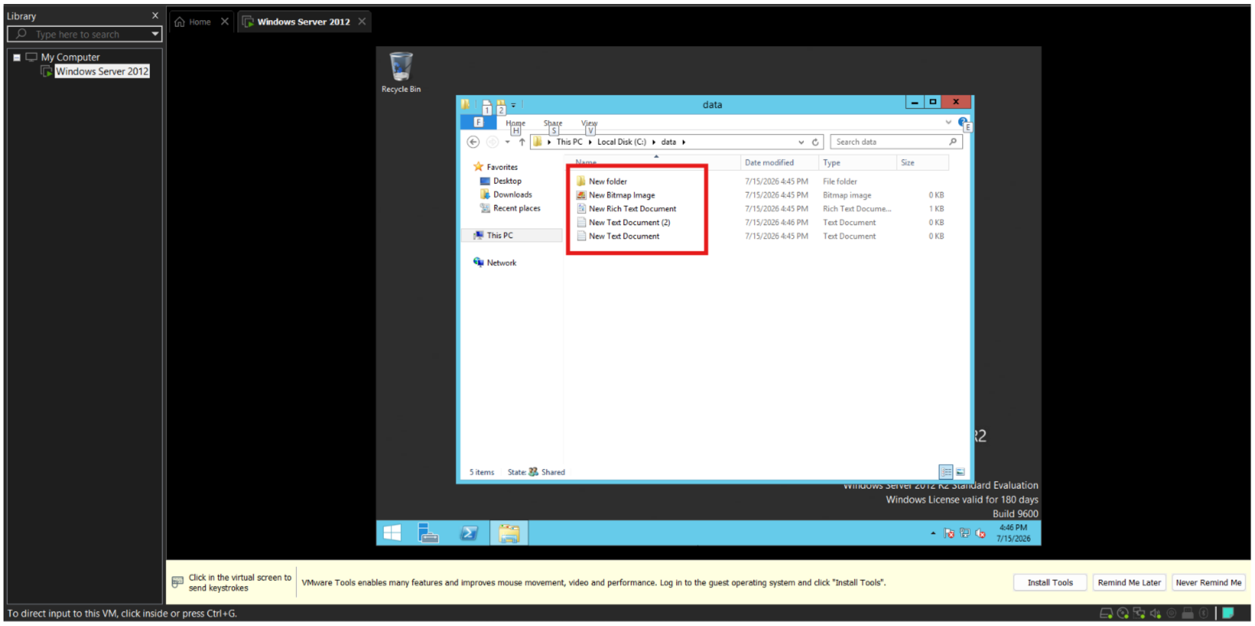
- Open the folder **Properties** by Right-clicking on it and go to sharing tab → Advanced Sharing and click on **Share this Folder**.



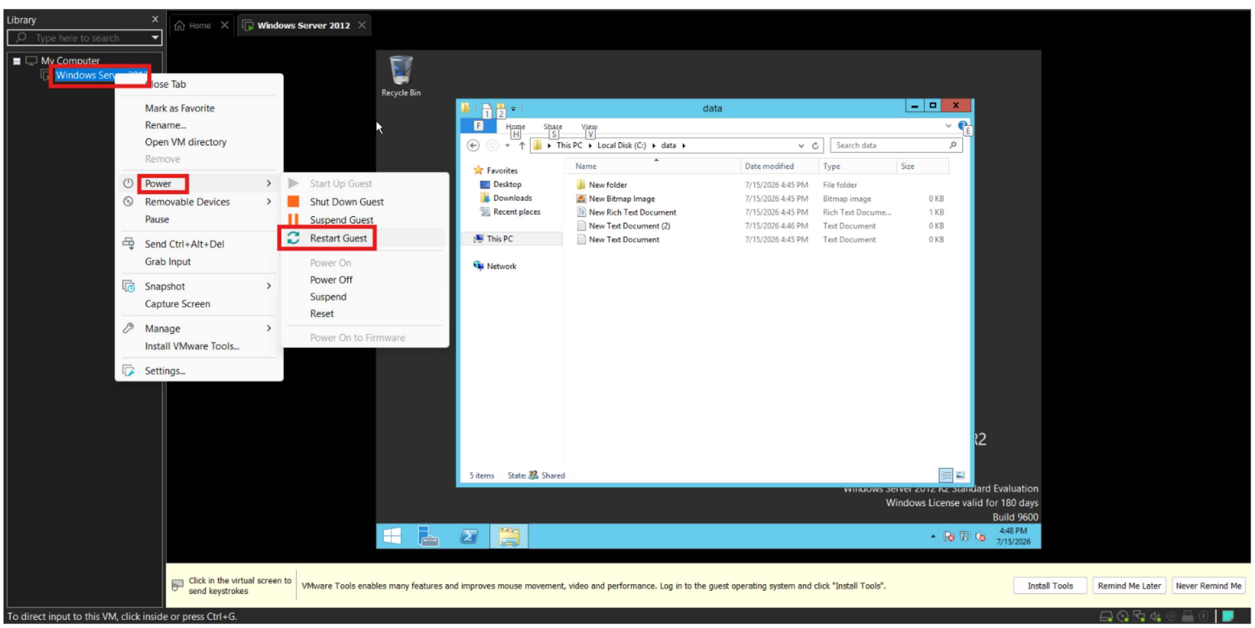
- Click on **Permissions** and give everyone **full control permissions** and click **Apply** and **OK**.



- Add a few sample files into the folder for transfer testing.

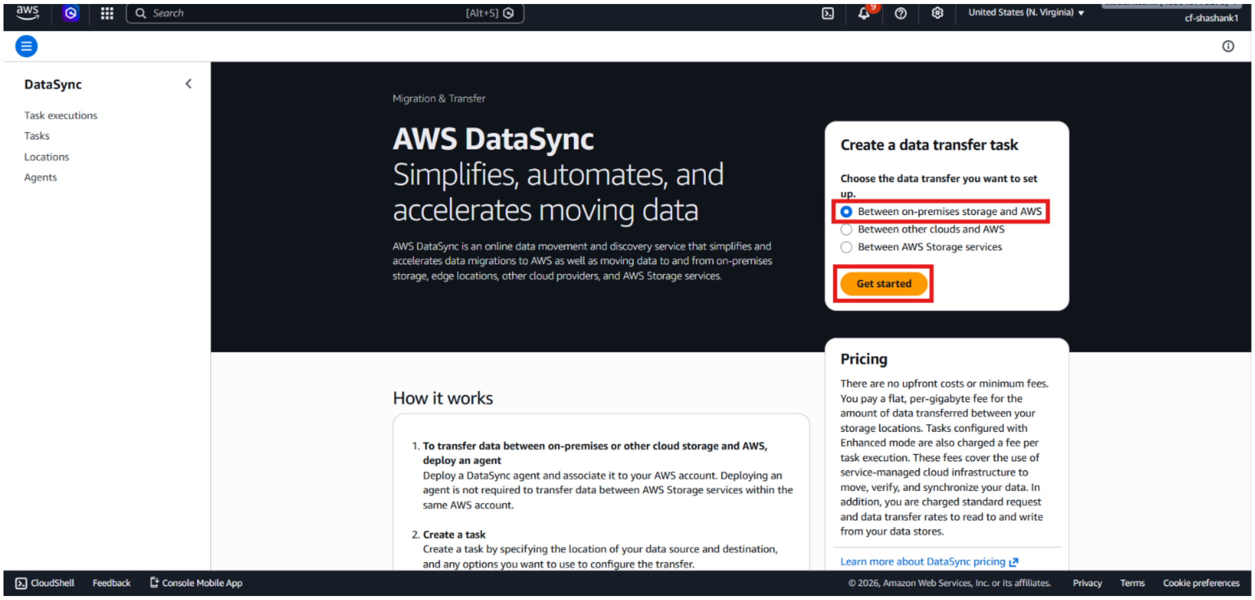


- And then restart the Virtual Machine by Right-clicking on Machine → Power → Restart Guest.

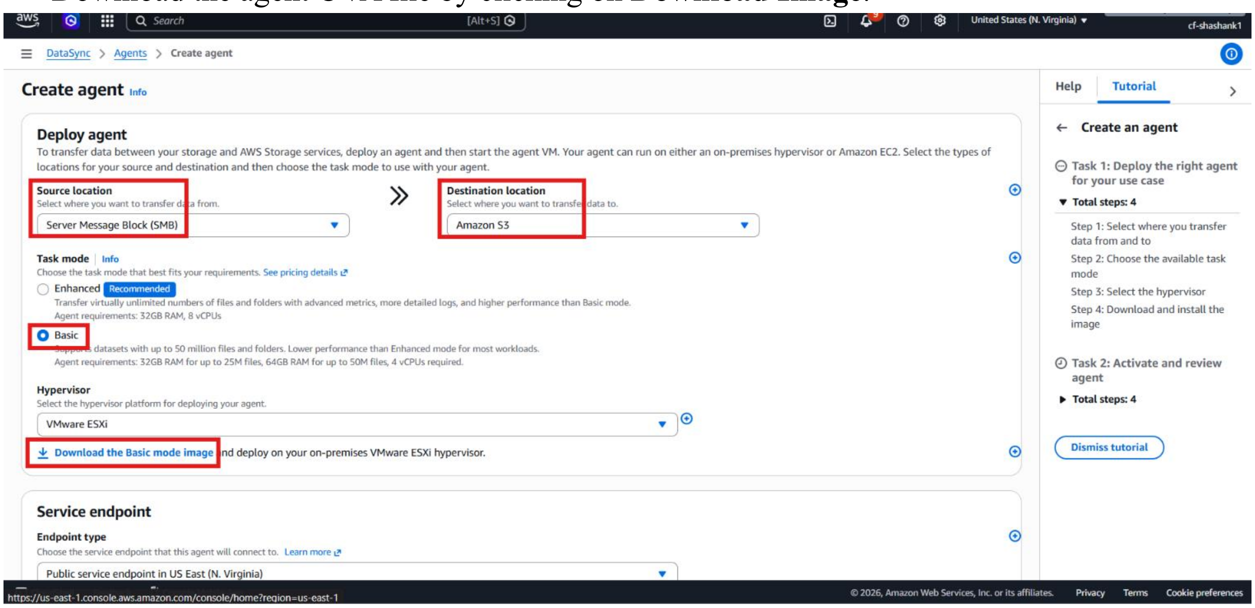


Step 4 – Download and import the DataSync agent

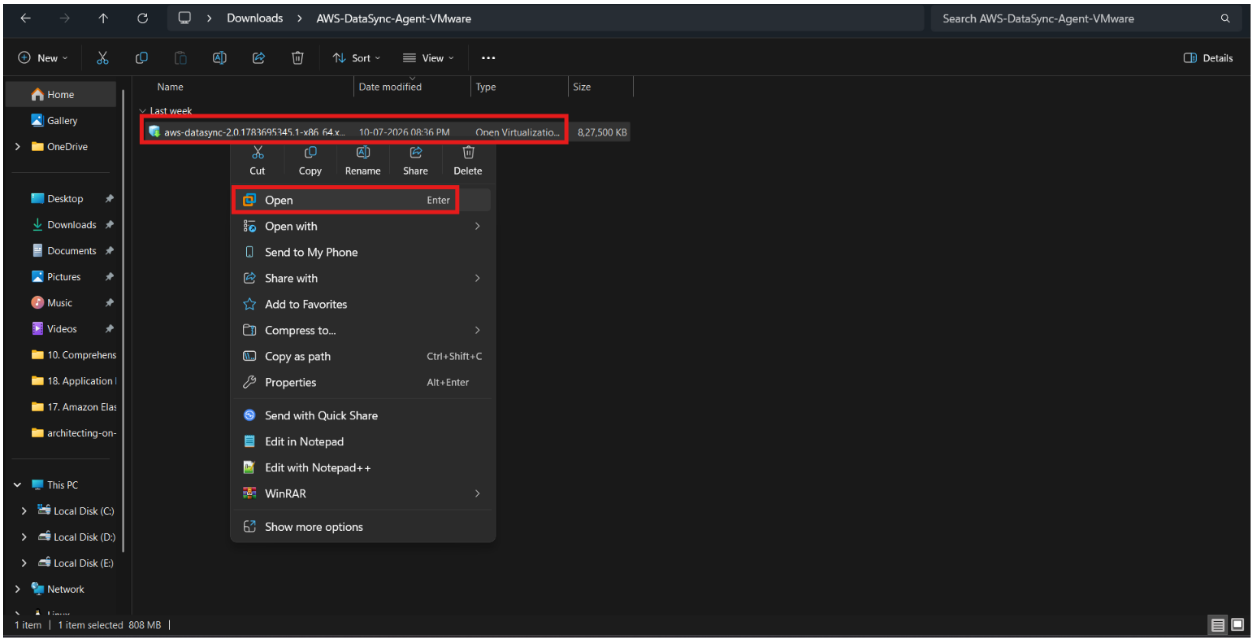
- Open the **AWS Management Console**.
- Navigate to **AWS DataSync**.
- Choose the option for transferring between on-premises storage and AWS.



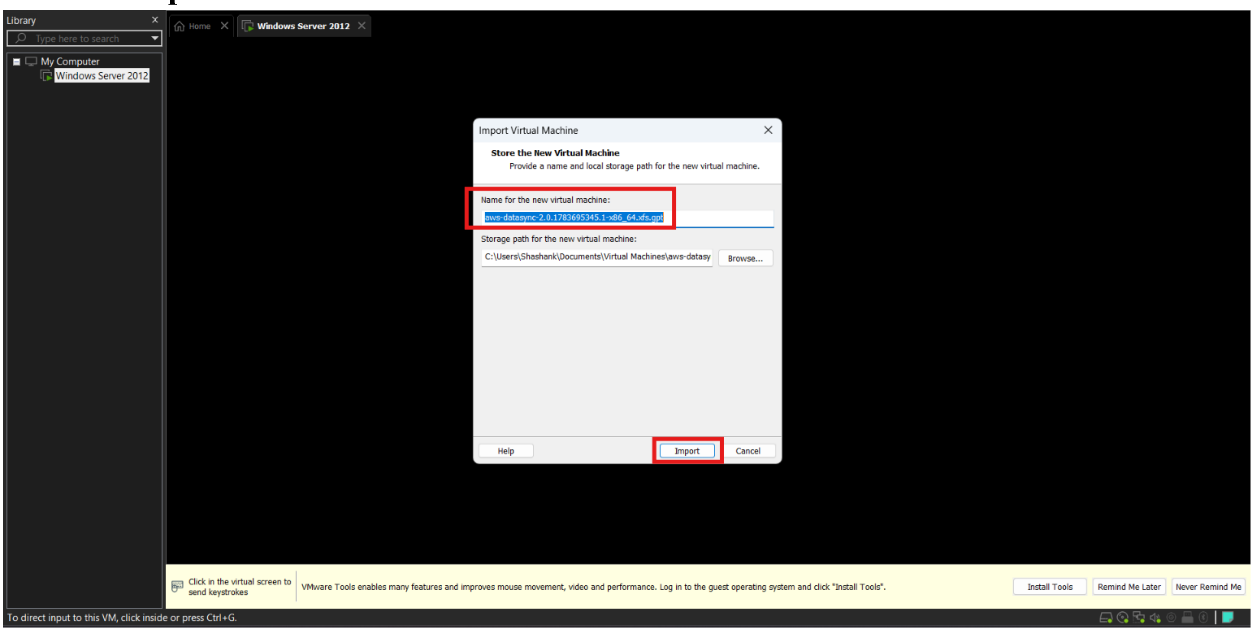
- Select Source and Destination location.
- Select the **VMware ESXi** agent image.
- Download the agent OVA file by clicking on **Download Image**.



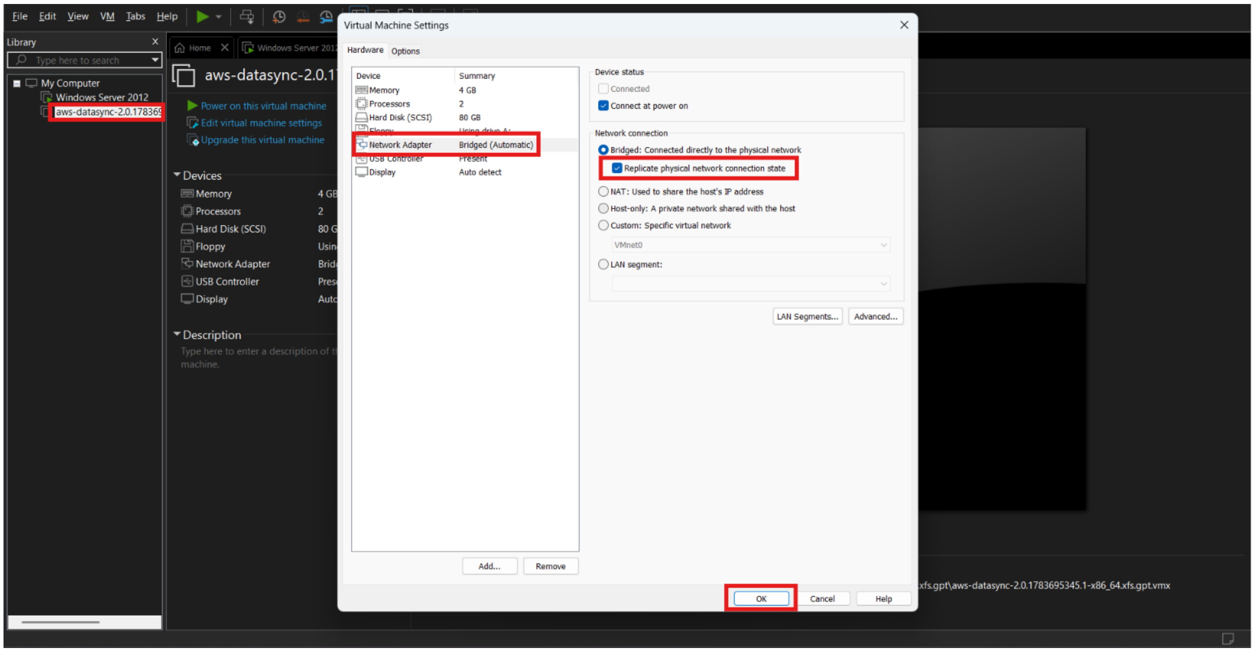
- Extract the zip folder into your computer and then open the extracted folder → **Right-click** on the **OVA file**.



- Click **Import**.



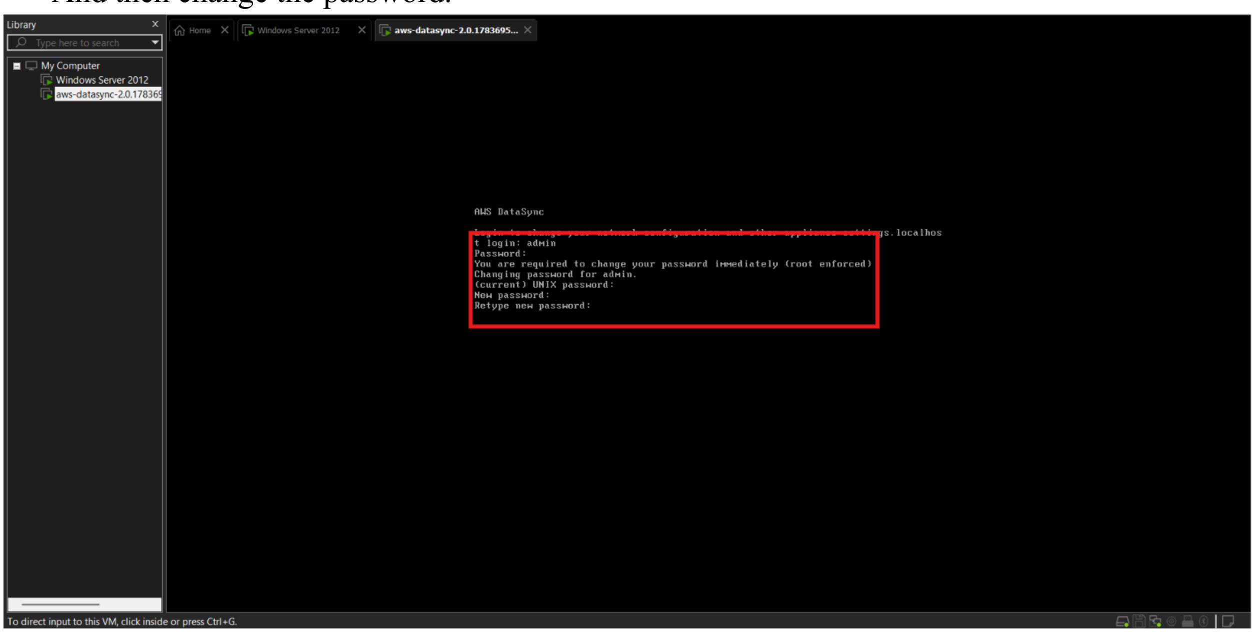
- Enable the same **Replicate physical network connection state** for this also.



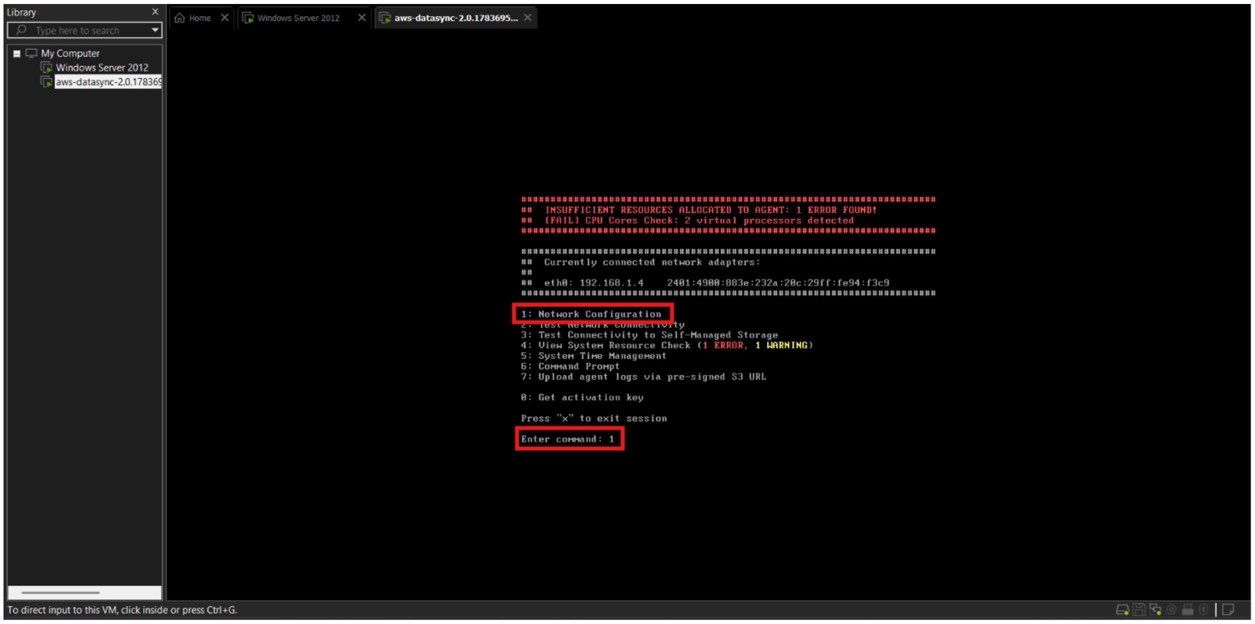
- Start the agent VM.

Step 5 – Configure the DataSync agent network

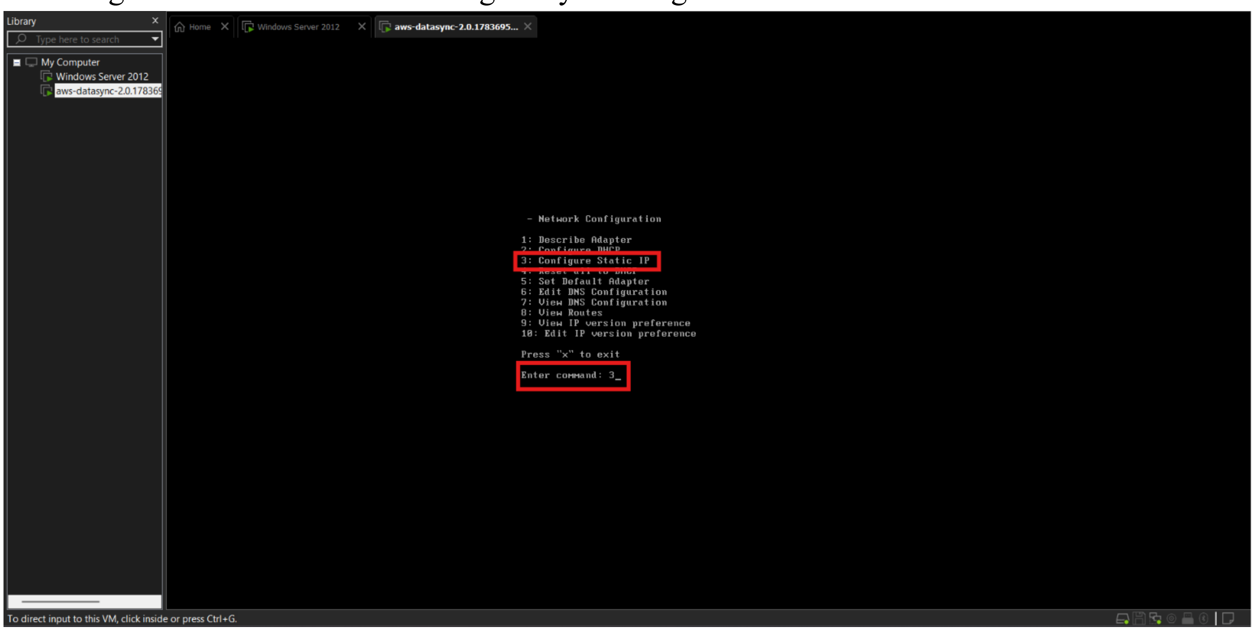
- Log in to the agent with the default admin credentials.
- Username – admin
- Password – password
- And then change the password.



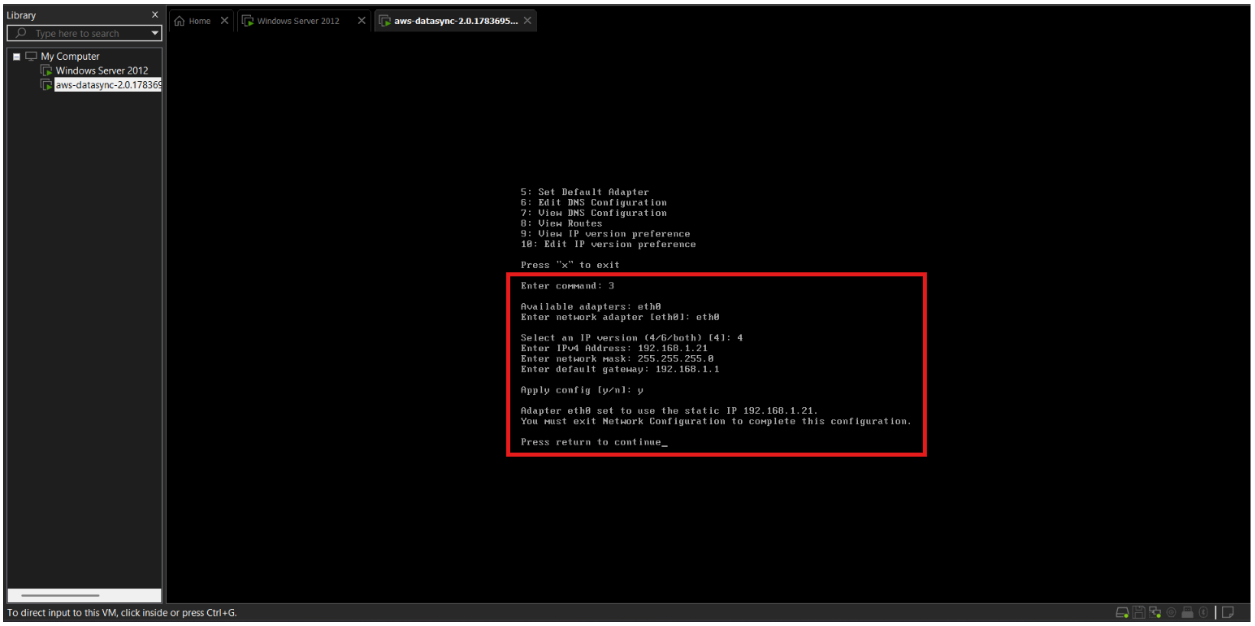
- Open the network configuration menu by entering 1.



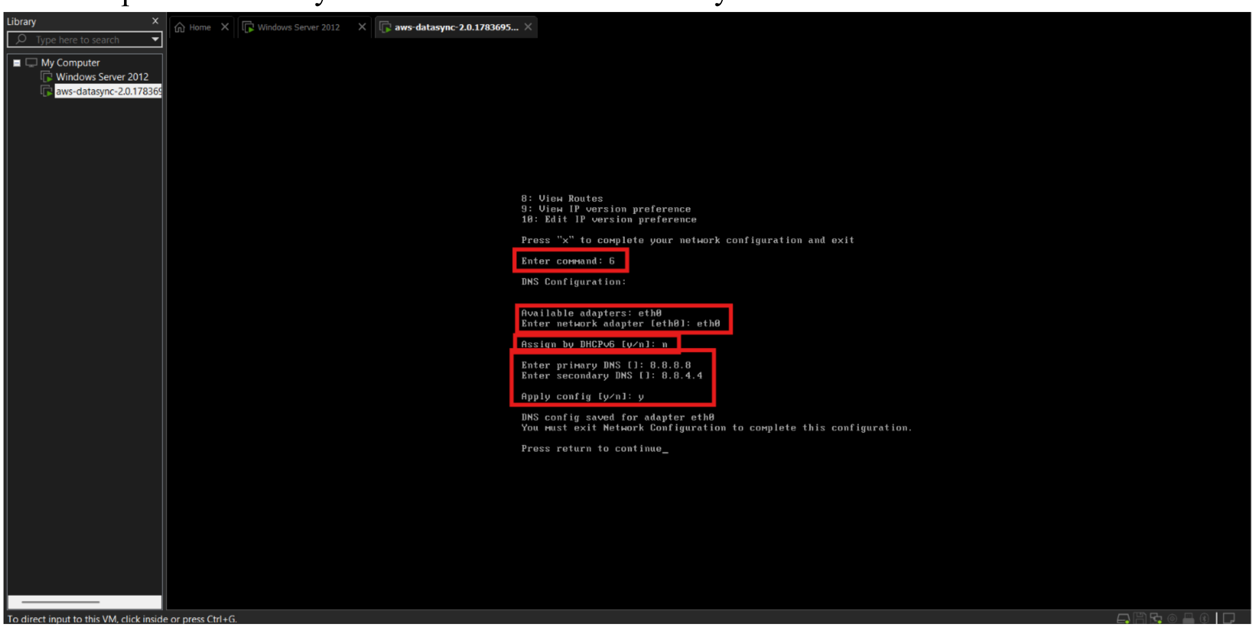
- Assign a static IP address to the agent by entering 3.



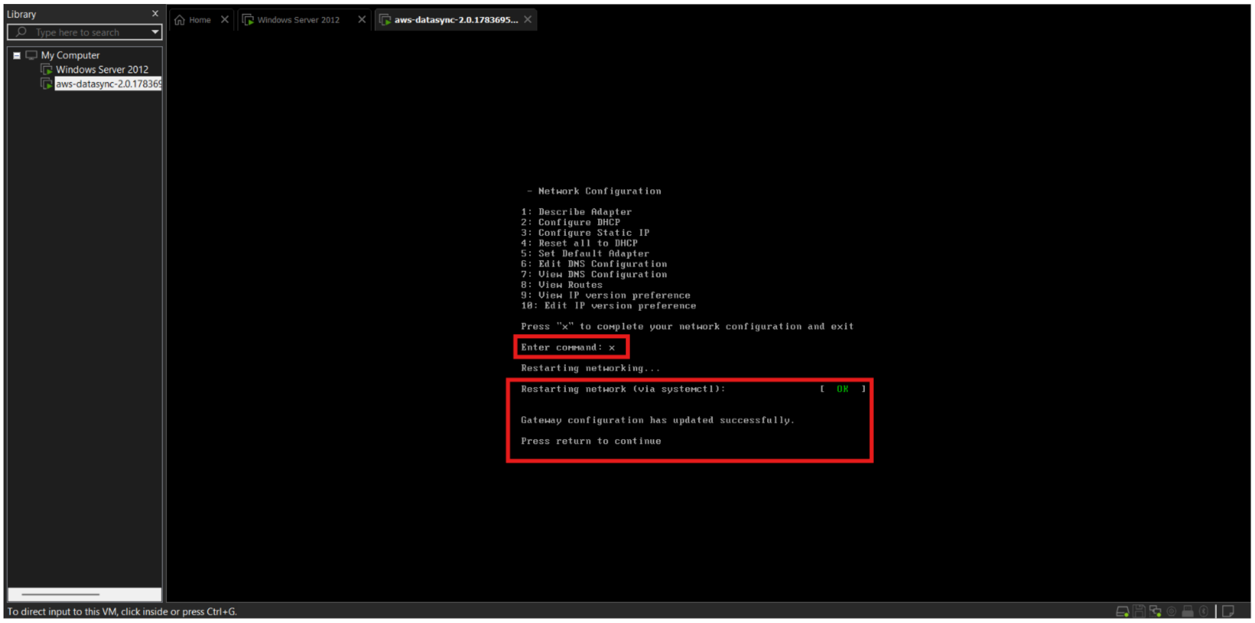
- Use the same set of IP as we used for configuring the Virtual Machine.
- Use the same subnet as the storage server.
- Configure the DNS server settings.



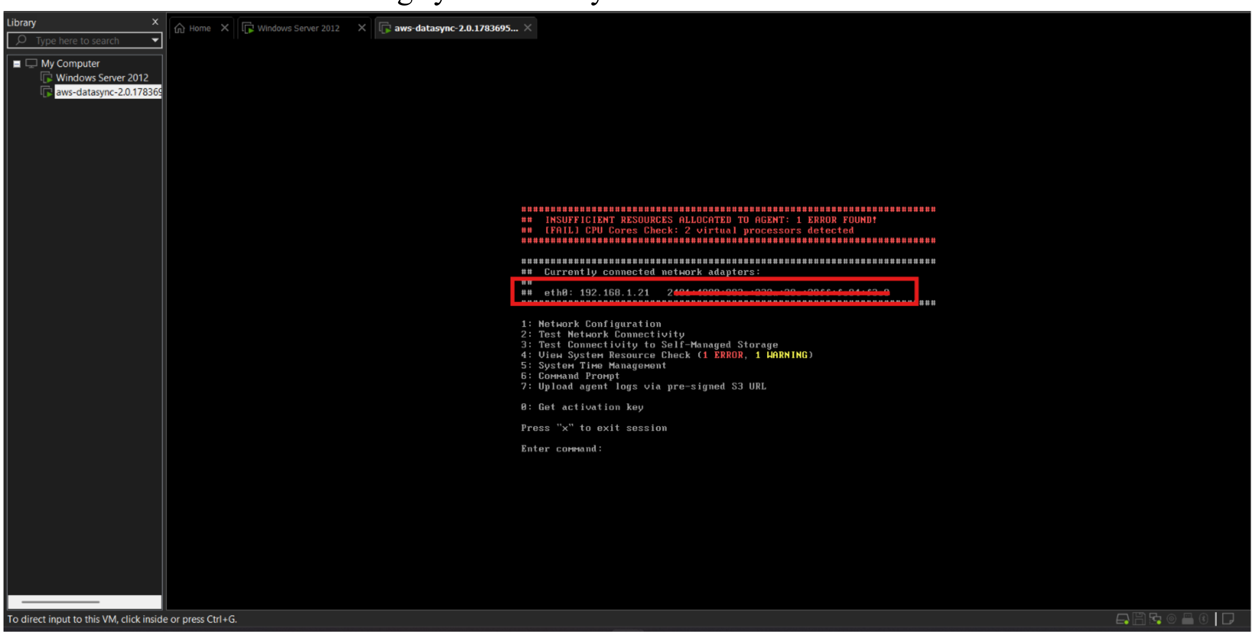
- Similarly, we have to configure the DNS Settings, for that enter 6.
- Now provide Primary DNS as 8.8.8.8 and Secondary DNS as 8.8.4.4



- Save the network configuration by pressing X.

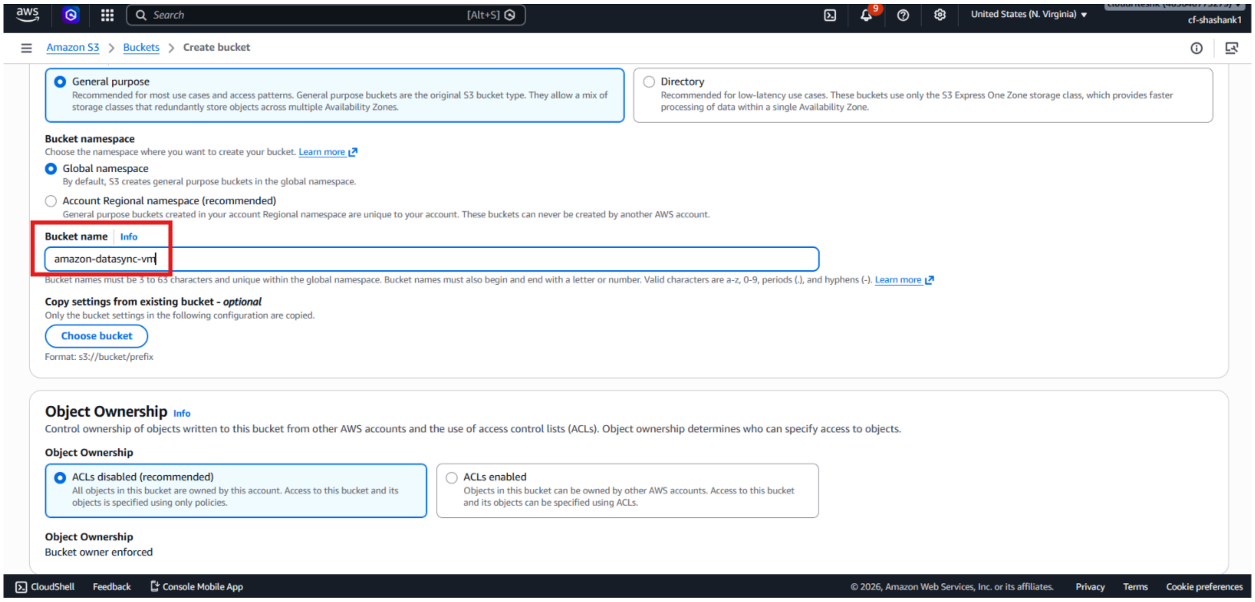


- Here after successful settings you can see your IP Address.



Step 6 – Create the S3 bucket

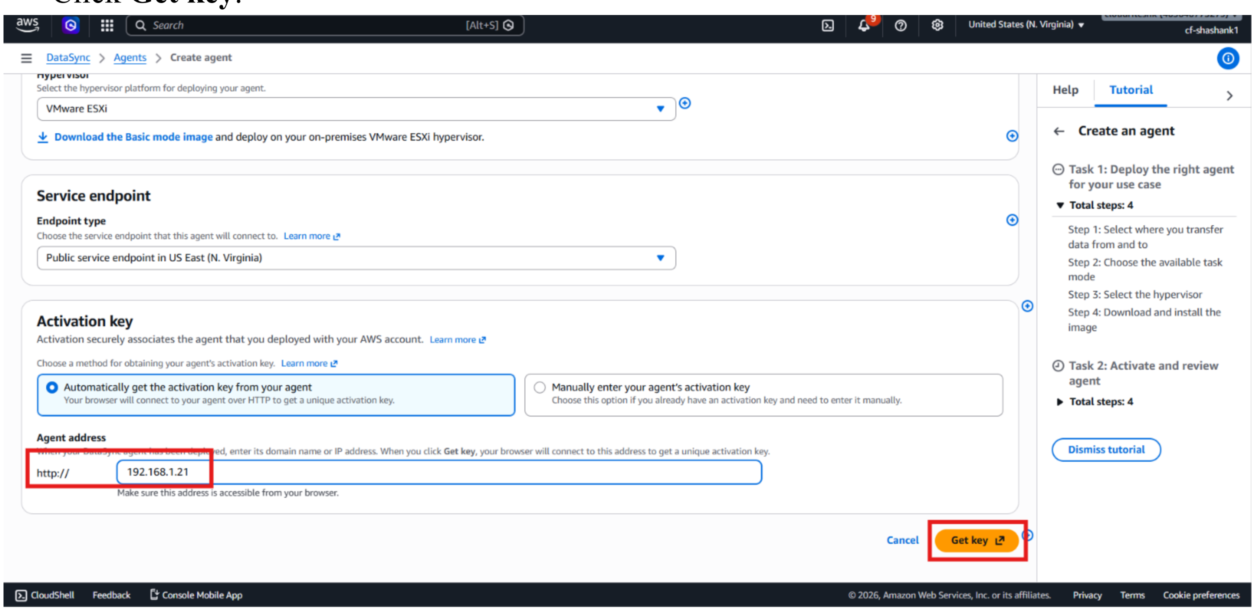
- Open **Amazon S3**.
- Create a new bucket named: **amazon-datasync-vm**.



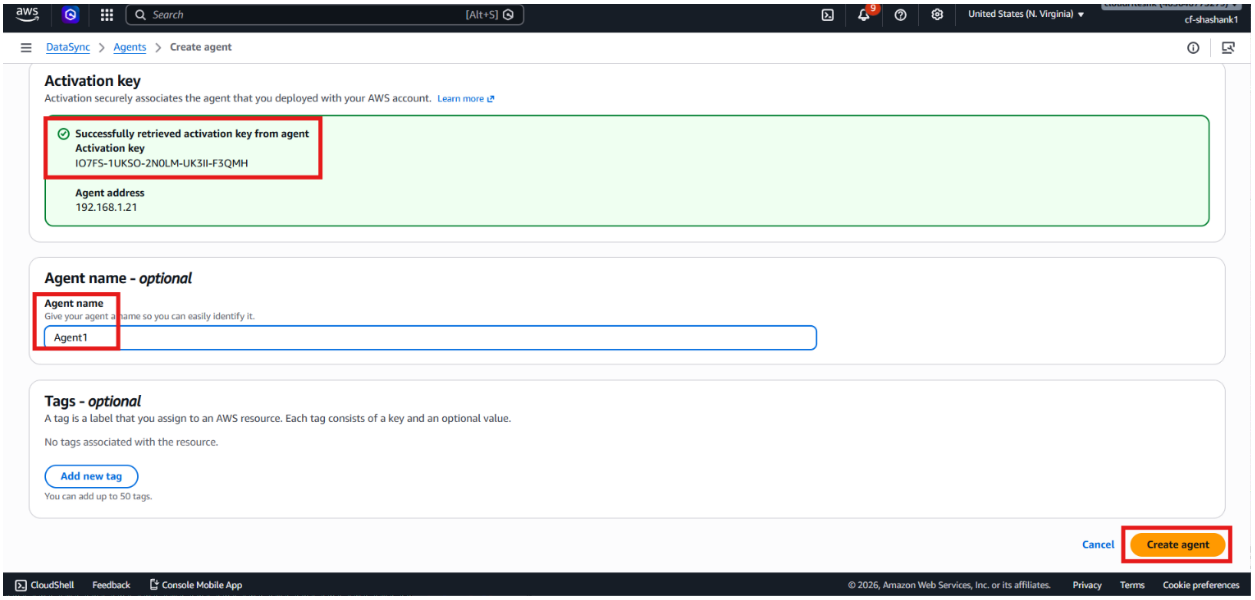
- Confirm that the bucket is empty before the transfer.

Step 7 – Activate the DataSync agent

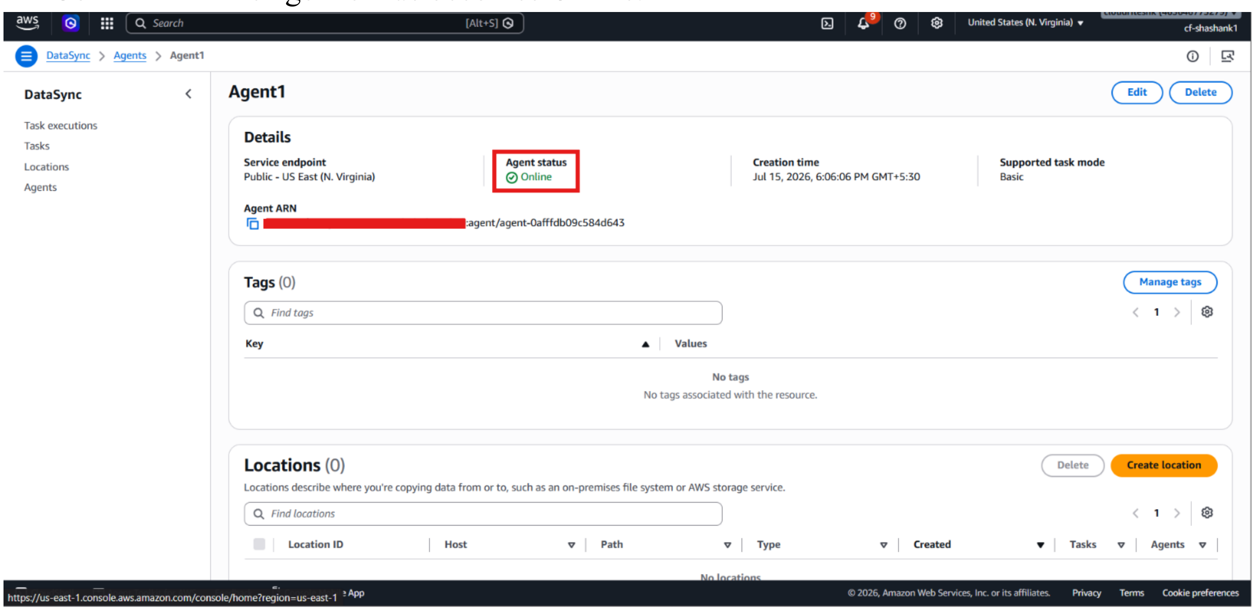
- Return to **AWS DataSync**.
- Choose the **on-premises storage and AWS** option.
- Enter the private IP address of the DataSync agent.
- Click **Get key**.



- Create and name the agent as **Agent1.**

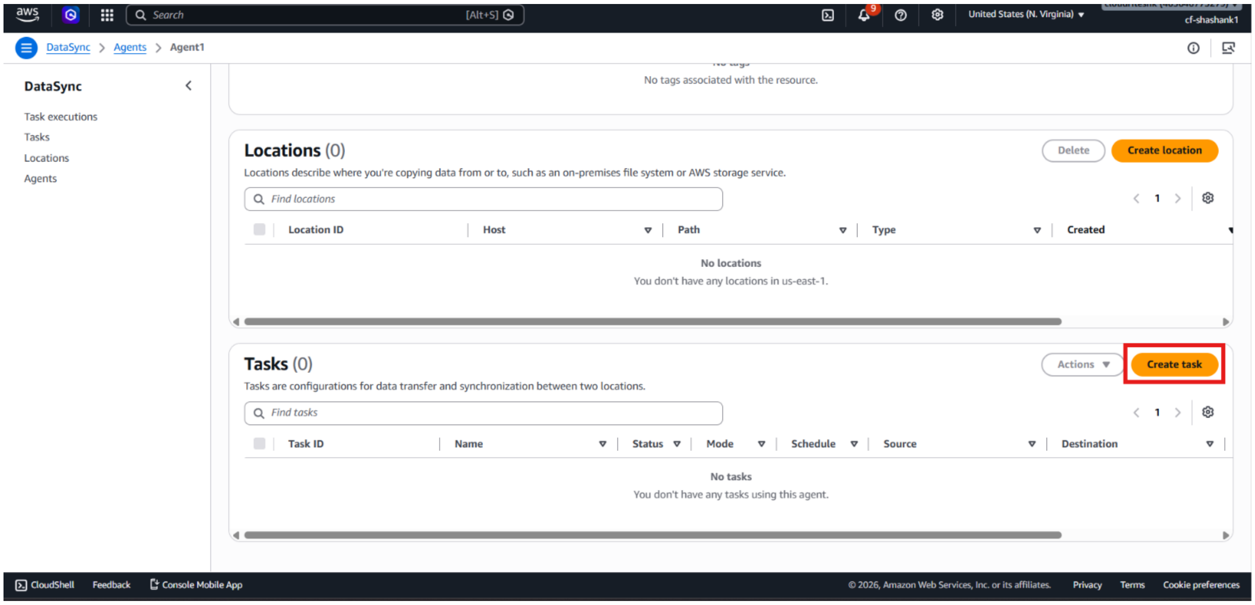


- Confirm that the agent status becomes Online.

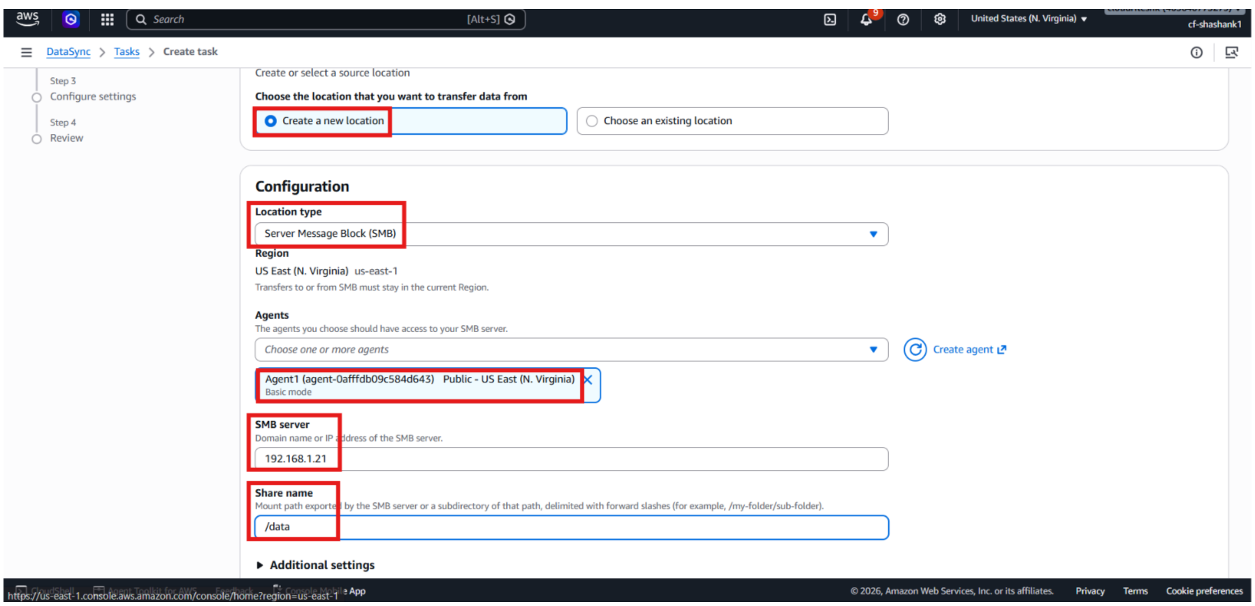


Step 8 – Create the DataSync task

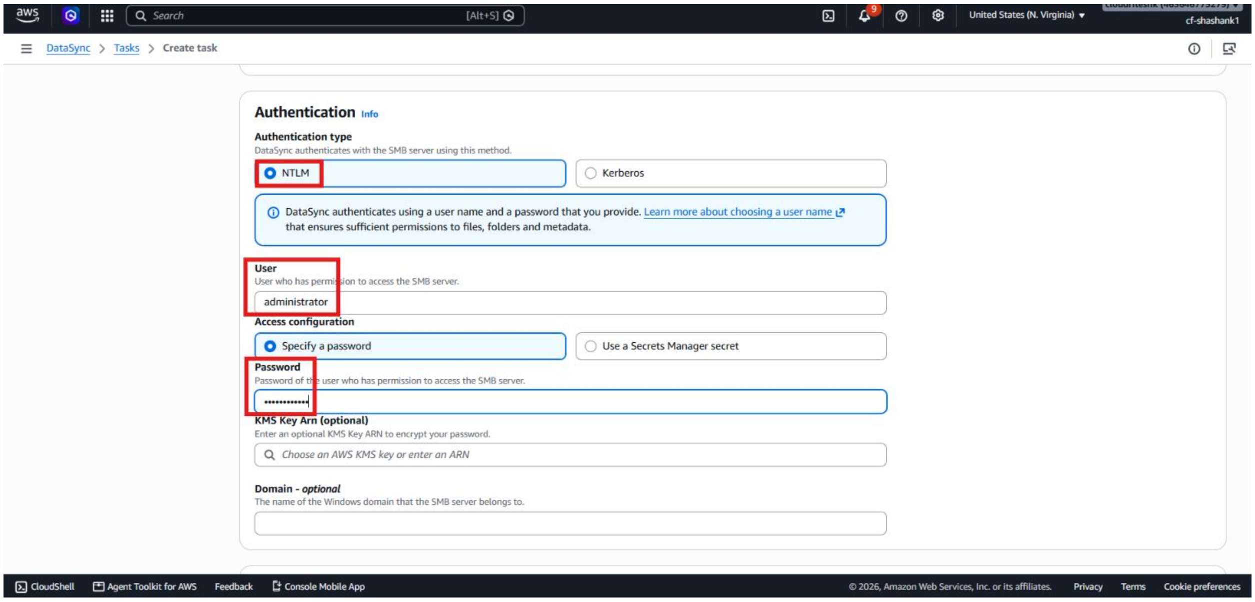
- Scroll down and click **Create task**.



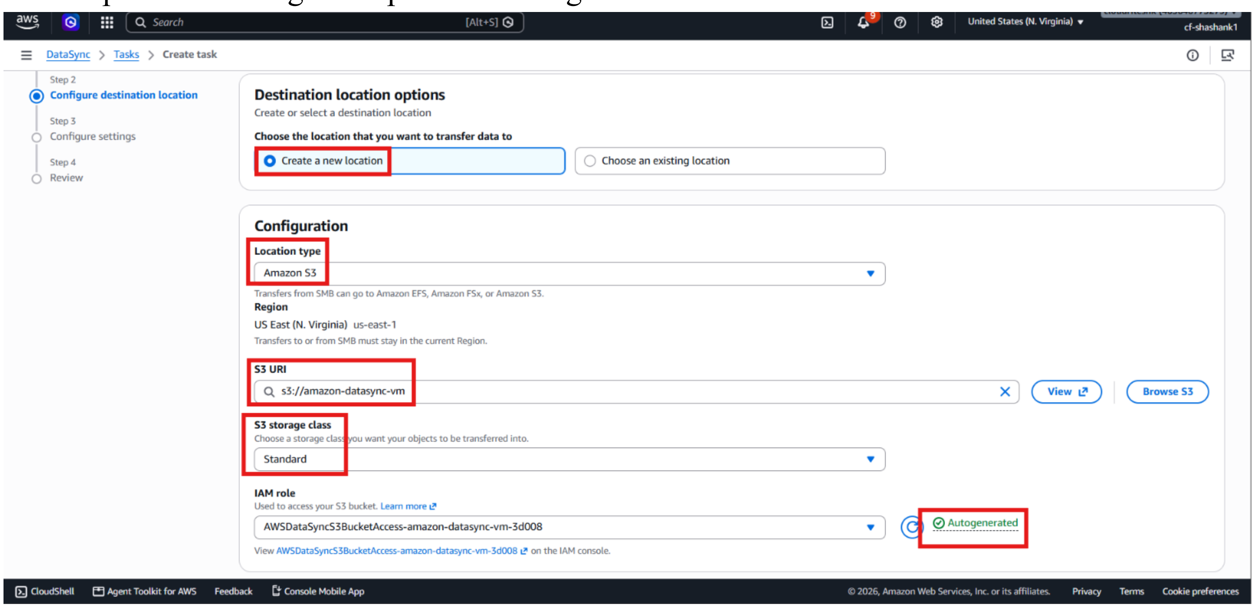
- Choose **SMB** as the source location.
- Enter the SMB server IP address.
- Choose the agent we have just created.
- Enter the share name data.



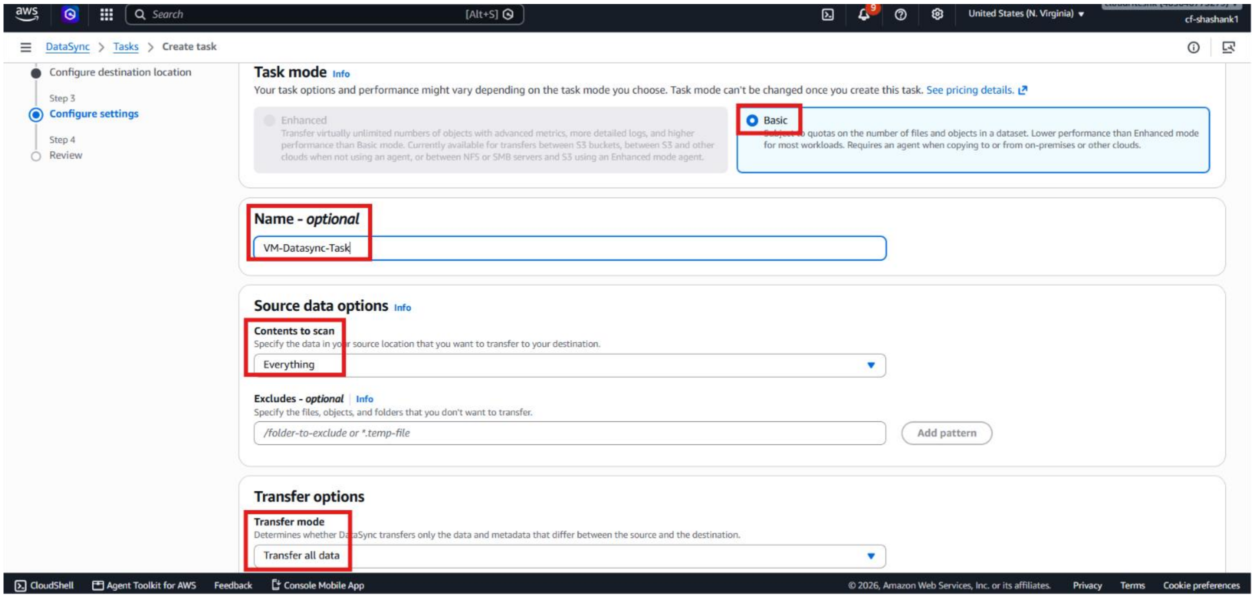
- Use the administrator username and password we have configured earlier while creating the Virtual Machine.



- Choose the S3 bucket **amazon-datasync-vm** as the destination.
- Let DataSync auto-generate the IAM role by clicking on the **Generate** button.
- Keep the remaining task options unchanged.



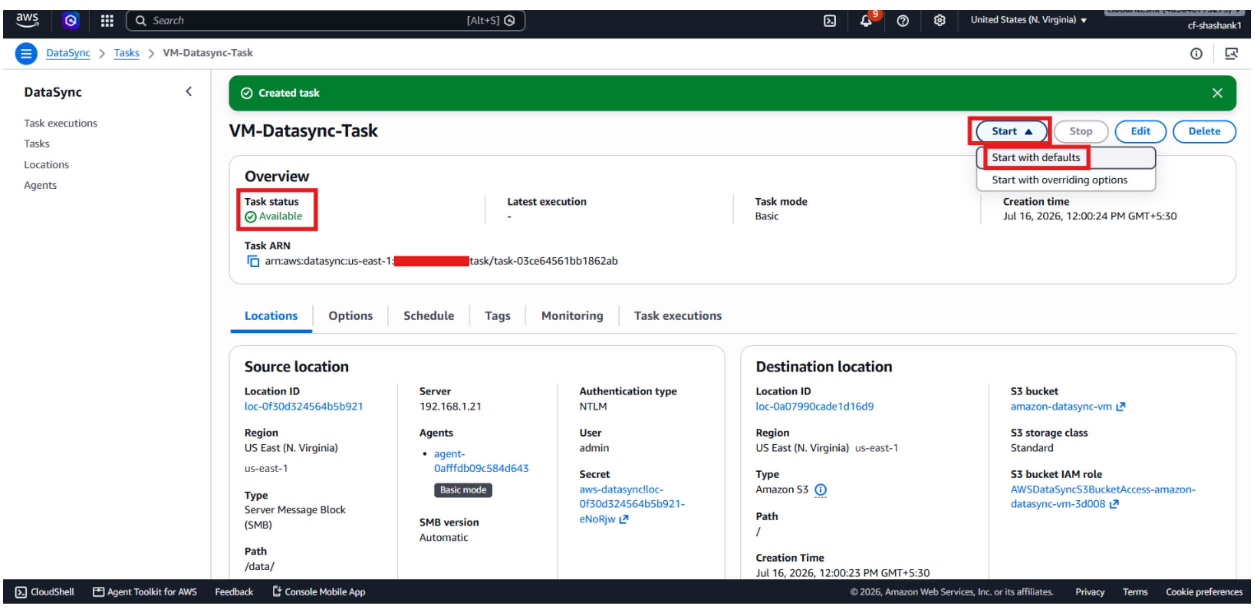
- Enter the name for the Task.
- Choose **Everything** for the Contents to Scan option.
- Set Transfer mode to **Transfer all data**.
- Leave rest settings as default.



- Create the task.

Step 9 – Start the transfer

- Open the created task.
- Click **Start → Start with defaults**.

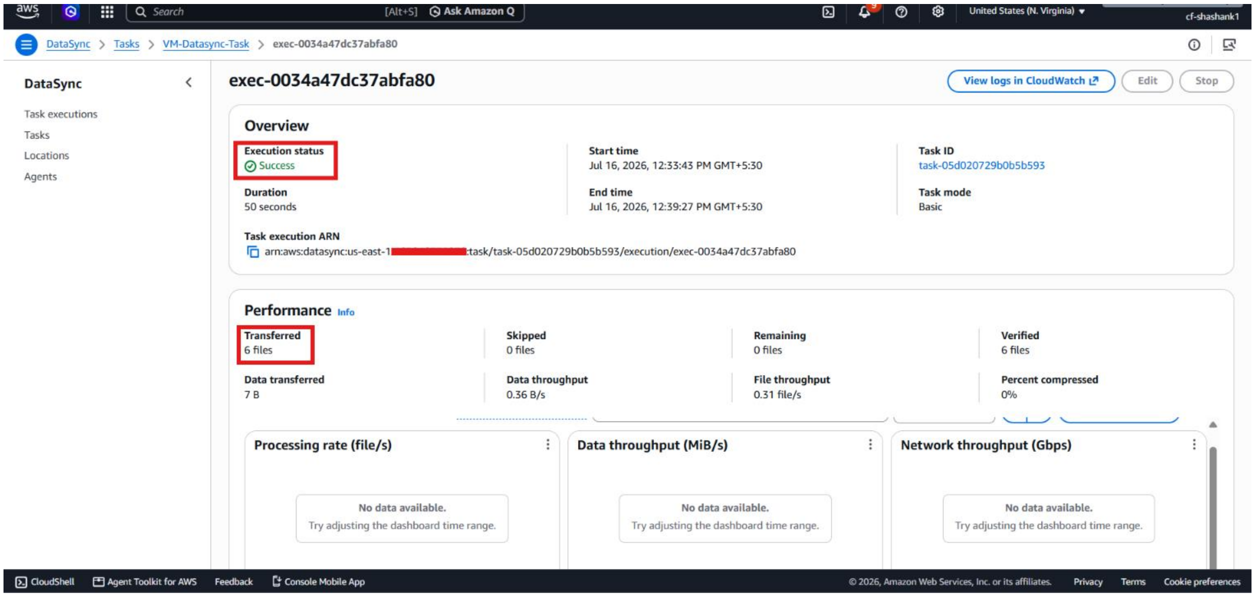


- Keep the default execution settings.
- Wait for the transfer to complete.

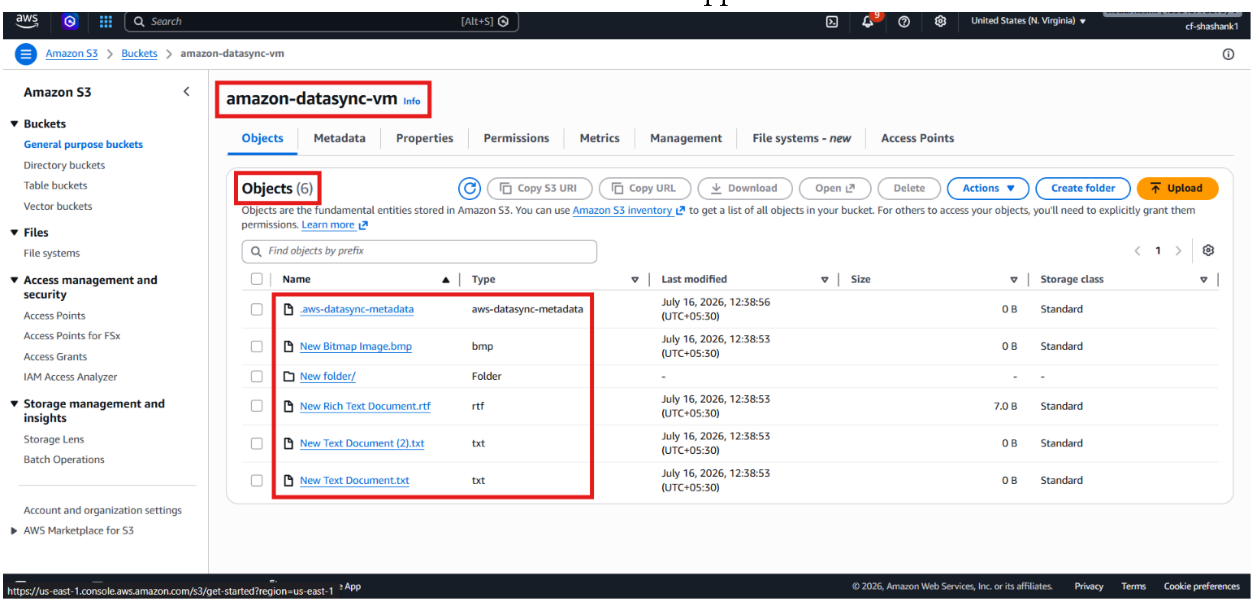
Step 10 – Verify the copied files

- Open the task history.
- Confirm that the transfer completed successfully.

- Check how many files were copied.



- Open the S3 bucket.
- Refresh the bucket contents.
- Confirm that the files from the shared folder now appear in **Amazon S3**.



Verification

- Verified that the Windows Server virtual machine was created successfully in **VMware Workstation**.
- Verified that the shared folder data was created and populated with sample files.
- Verified that the **AWS DataSync** agent was downloaded, imported, and activated successfully.
- Verified that the SMB source location pointed to the correct storage server and share.

- Verified that the destination **Amazon S3** bucket on prem data was created successfully.
- Verified that the DataSync task completed successfully.
- Verified that the files were copied from the SMB share into **Amazon S3**.